

1 Person, Labor, User, & Craft

1.1 Overview

Person represent an individual that related to EMS system. Person record is created for any individual whose name appears anywhere on any data. Person record is required when creating a user or a labor. A person data contain basic information about an individual's name, address, contact information, and other generic information.

Labor represent employees and contractors who perform work on work orders. Labor records contain information about an individual's skills and qualifications. These records are used to plan and schedule work, and to track labor costs for work orders. A labor record must have a person record associated with it.

User represent authentication to access the system. User records contain user names, passwords, and security profiles that determine which applications, options, and data a user can access. A user must have a person record. A labor record and user record can be associated to the same person record.

Crafts represents an occupation or trade, and typically the craft name reflects the type of work done by members of the trade. If pay rates differ based on expertise, multiple skill levels with different pay rates for each skill level can be assigned to the craft. For each craft user can specify standard rates, and add premium pay codes for premium pay rates.

1.2 People Application

People application is used to create, modify, view, and delete records person data. The **People** application stores personal information, employee information, and delegation configurations.

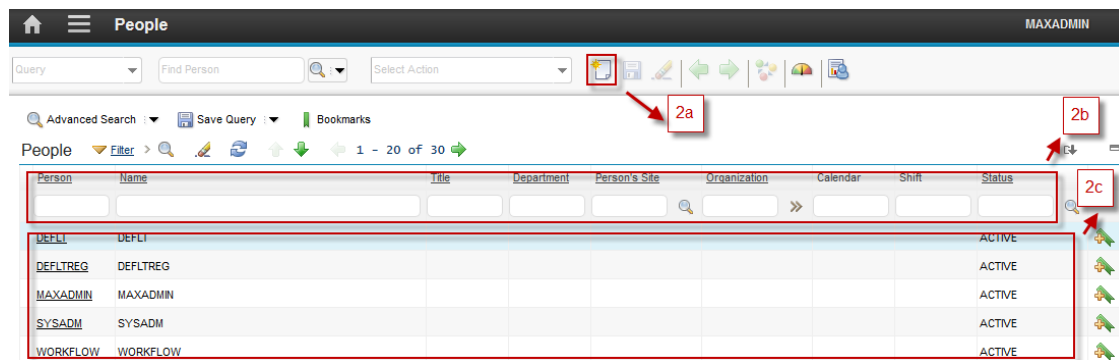
Followings are details about **People** applications accessibility:

1



From **Go To** menu, choose **Administration – Resources – People**

2



Header Bar will give information that user is in the **People** application.

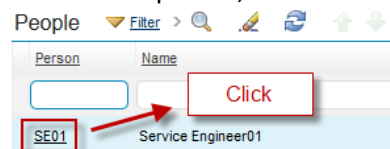
2.a **New Person** button, to add new person

2.b **Filter** person by **Person ID, Name, Title, Status**, etc.

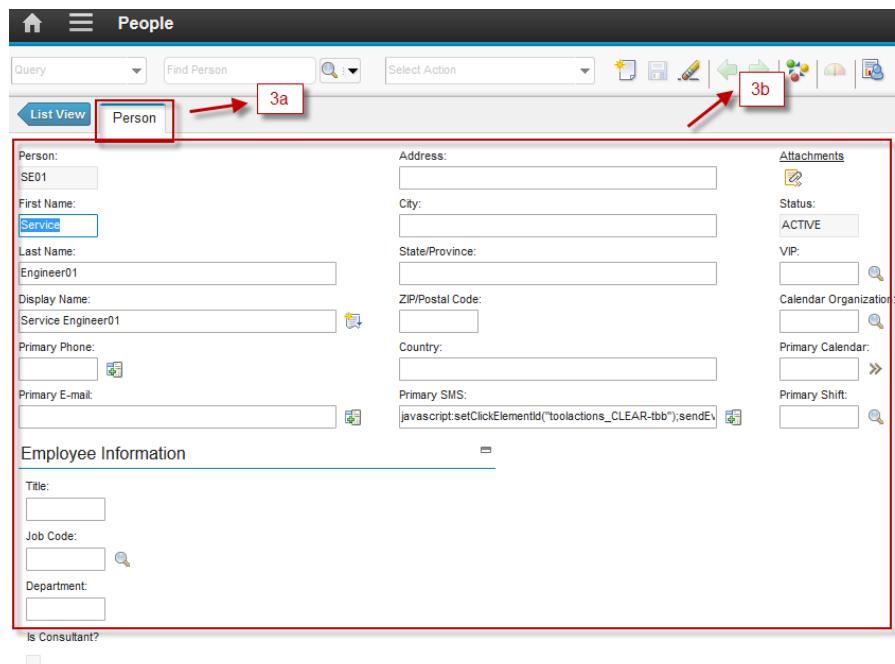
2.c List of all persons based of filtering result.

3

Choose one person, to show the details of the person

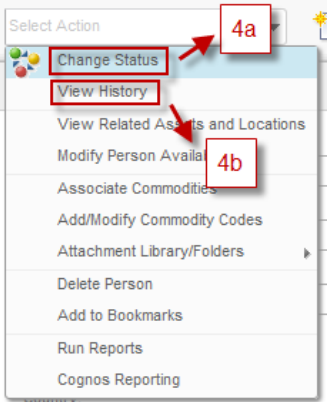


Will show display like below picture.



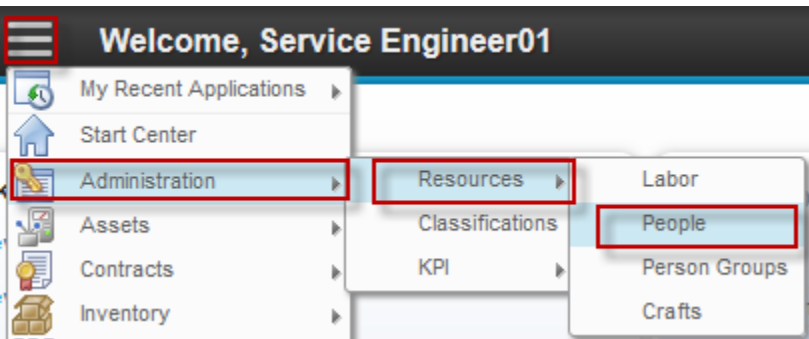
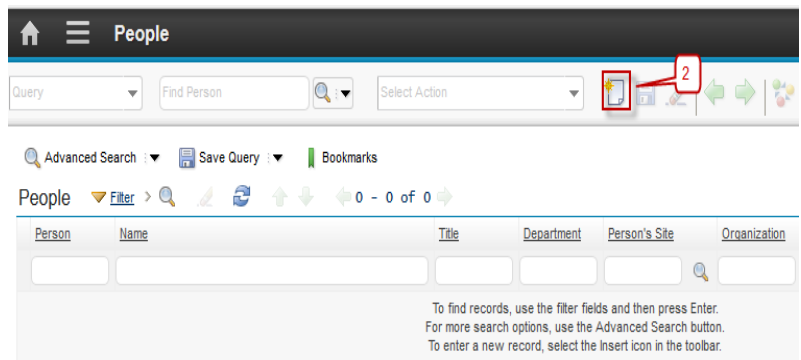
3.a **Person** tab, to view and update details of person

3.b Detail information of the person, such as first name, last name, display name, primary phone, primary email, etc.

4	User can perform some action related to the person from Select Action menu (on top of the form).  4.a Change status action to change status this person 4.b View History, to view the change status history of the person.
---	---

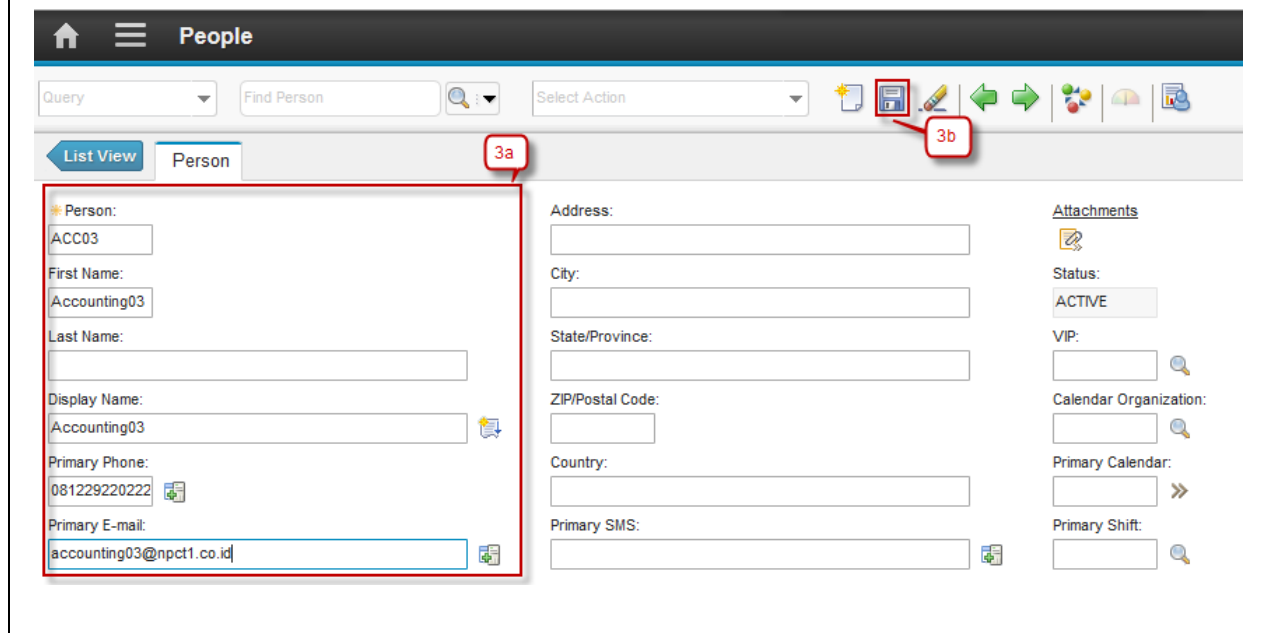
1.2.1 Create Persons

Person creation data can be created by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create new person:

Step	Action
1. Go to People application by click  (GoTo) – Administration Resources – People .	
2. Click  New Person .	

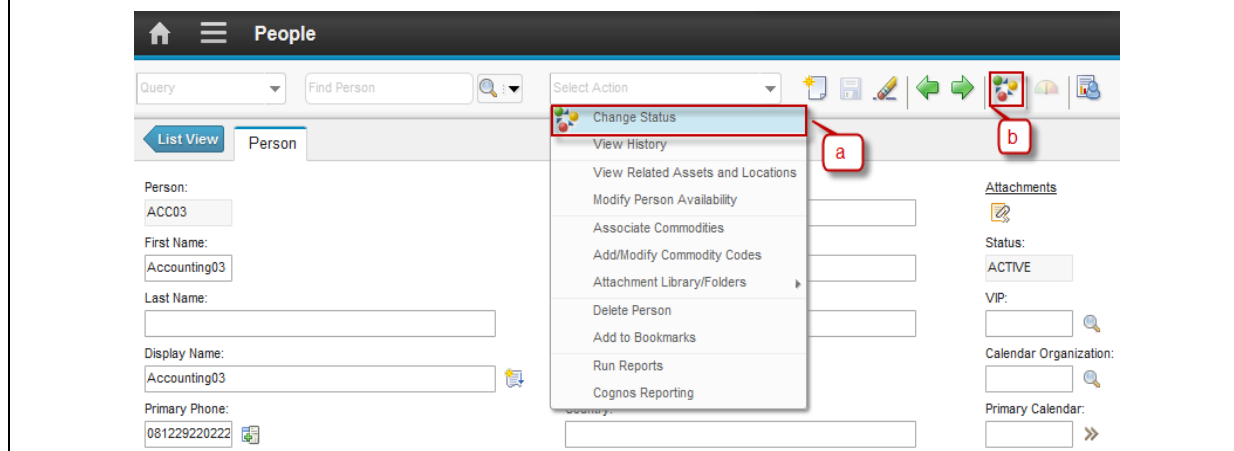
3a. Input the Person Code and any additional information about Person.

3b. Click **Save**.

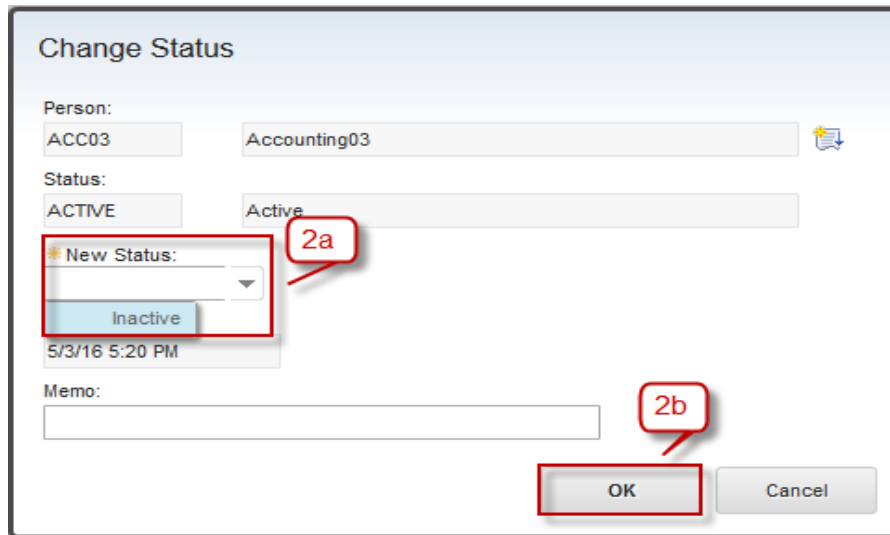


1.2.2 Change Status of Person

Person status change can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Change Status of a person data:

Step	Action
1. Change Status of Person have two ways:	
a. From Select Action → Change Status	
b. Click button 	
	

2a. Dialog box **Change Status** will be appeared, see the figure below :

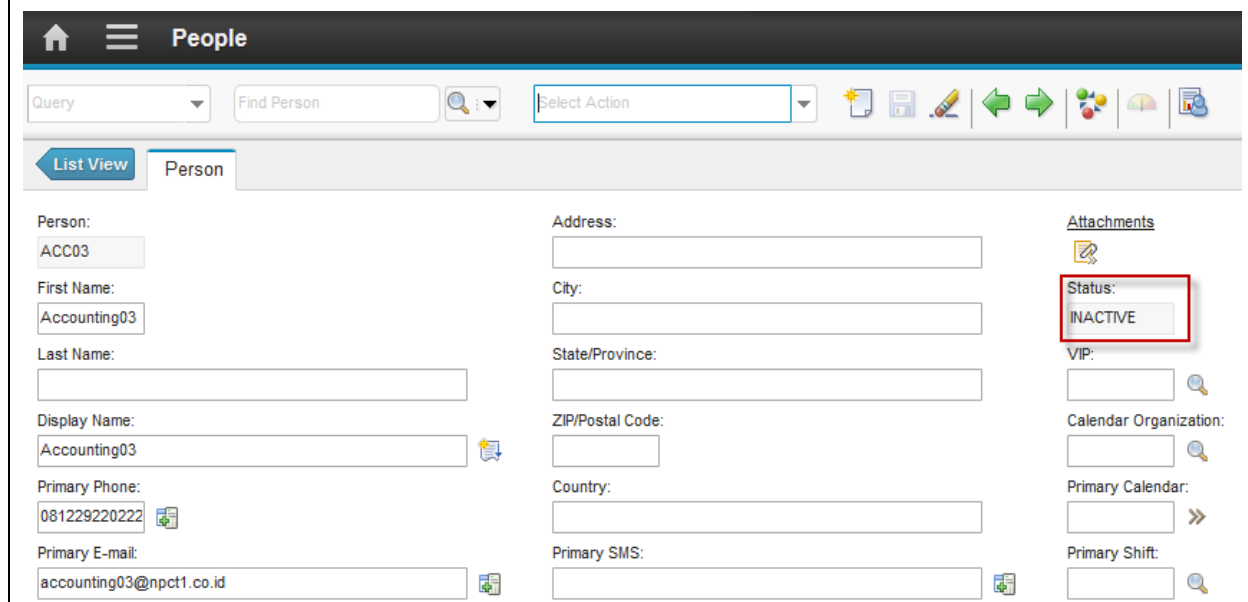


The 'Change Status' dialog box is shown. It contains the following fields and controls:

- Person:** ACC03 (selected in a dropdown), Accounting03 (text input)
- Status:** ACTIVE (selected in a dropdown), Active (text input)
- New Status:** A dropdown menu with 'Inactive' selected. A red callout box labeled '2a' points to this dropdown.
- Memo:** A text input field.
- Timestamp:** 5/3/16 5:20 PM
- Buttons:** OK and Cancel. A red callout box labeled '2b' points to the OK button.

2b. Click button **OK**

3. Status will be changed to **INACTIVE**

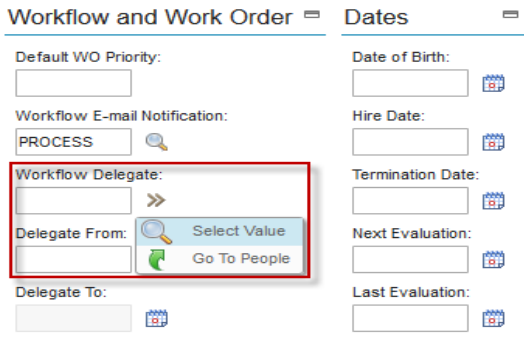
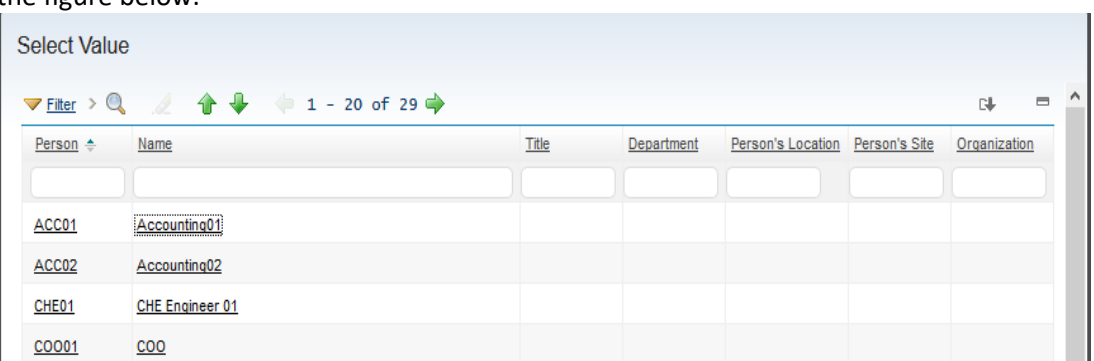
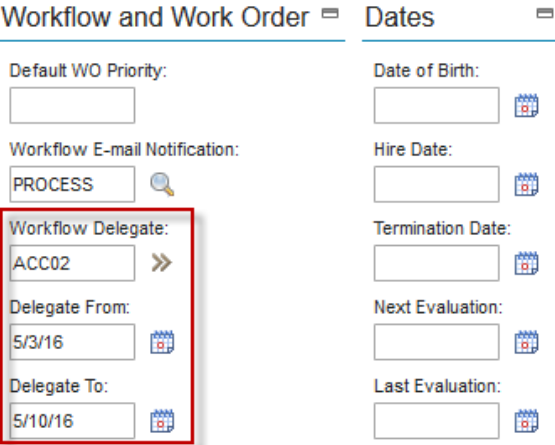


The 'People' management interface is shown. It includes a header bar with a home icon, a menu icon, and the title 'People'. Below the header is a search bar with 'Find Person' and a 'Select Action' dropdown. The main content area is divided into two tabs: 'List View' and 'Person'. The 'Person' tab is active, showing a form for a person with the following fields:

- Person:** ACC03
- First Name:** Accounting03
- Last Name:** (empty)
- Display Name:** Accounting03
- Primary Phone:** 081229220222
- Primary E-mail:** accounting03@npct1.co.id
- Address:** (empty)
- City:** (empty)
- State/Province:** (empty)
- ZIP/Postal Code:** (empty)
- Country:** (empty)
- Primary SMS:** (empty)
- Attachments:** A section with a red box around the 'Status' field, which is set to 'INACTIVE'. Other fields in this section include VIP, Calendar Organization, Primary Calendar, and Primary Shift.

1.2.3 Set Person's Delegation

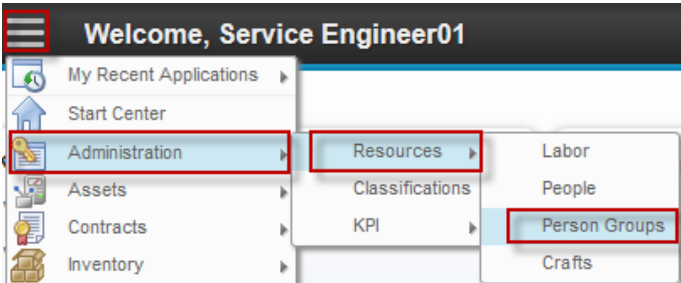
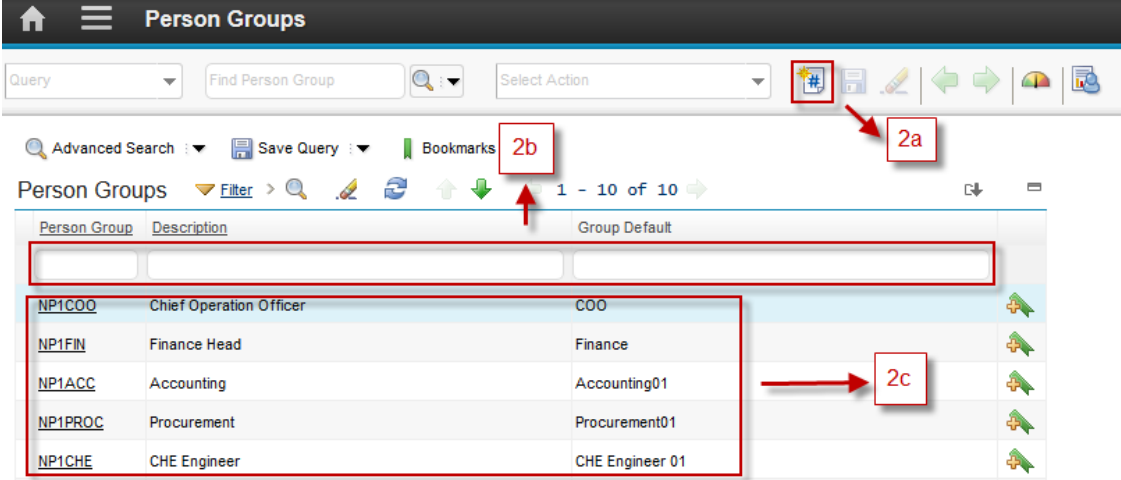
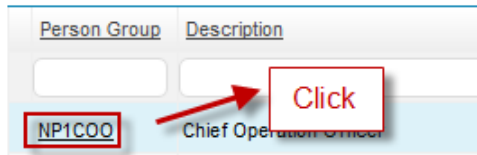
When a person is off, delegation can set toward those person, so any upcoming assignment can be redirected to the delegation. Person's delegation can be configured by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to set person's delegation:

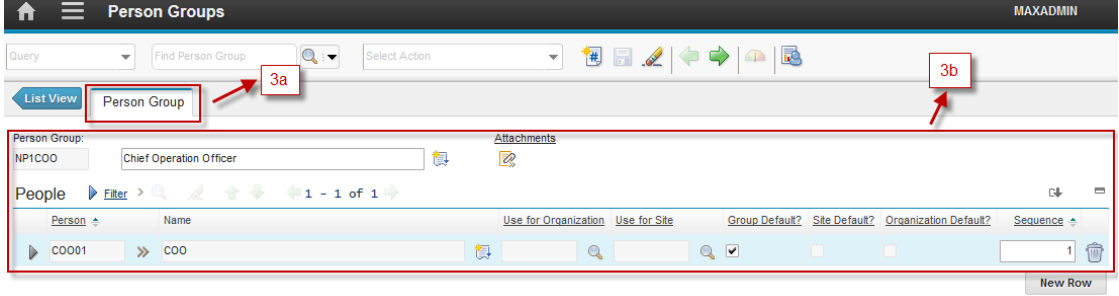
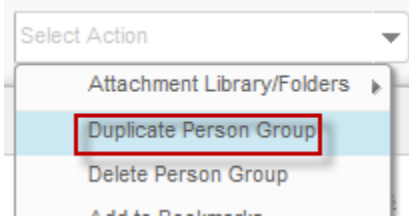
Step	Action
1. Go to section Workflow and Work Order → click button » → Select Value or Go To People to select the person who will be delegated	
2. When click button Select Value will be appeared list of all person and select one of record, see the figure below:	
3. Input field Delegate From and field Delegate To using button 	

1.3 Person Groups Application

Person Groups application is used to manage the groups of person within organization. A person group can be the recipient of a document that is routed by a workflow process. If a document is routed to a person group, everyone in the group receives the document, unless it is configured to send the document only to the person whose calendar indicates availability.

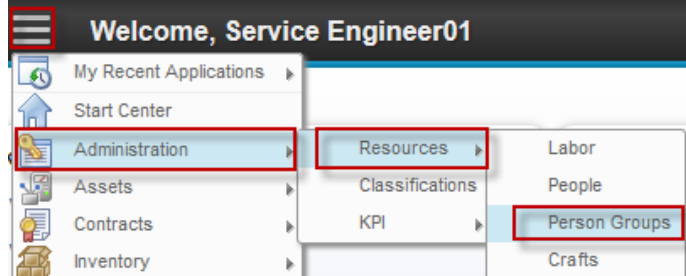
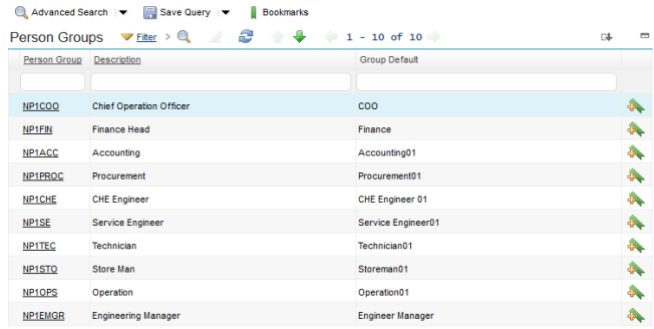
Followings are details about **Person Groups** applications accessibility:

1	 From Go To menu, choose Administration – Resources – Person Groups																		
2	 Header Bar will give information that user is in the Person Group application. 2.a New Person Group button, to add new person 2.b Filter person group by Person Group Code, Description, etc. 2.c List of all person group based of filtering result. <table border="1" data-bbox="272 1100 1284 1360"> <thead> <tr> <th>Person Group</th> <th>Description</th> <th>Group Default</th> </tr> </thead> <tbody> <tr> <td>NP1COO</td> <td>Chief Operation Officer</td> <td>COO</td> </tr> <tr> <td>NP1FIN</td> <td>Finance Head</td> <td>Finance</td> </tr> <tr> <td>NP1ACC</td> <td>Accounting</td> <td>Accounting01</td> </tr> <tr> <td>NP1PROC</td> <td>Procurement</td> <td>Procurement01</td> </tr> <tr> <td>NP1CHE</td> <td>CHE Engineer</td> <td>CHE Engineer 01</td> </tr> </tbody> </table>	Person Group	Description	Group Default	NP1COO	Chief Operation Officer	COO	NP1FIN	Finance Head	Finance	NP1ACC	Accounting	Accounting01	NP1PROC	Procurement	Procurement01	NP1CHE	CHE Engineer	CHE Engineer 01
Person Group	Description	Group Default																	
NP1COO	Chief Operation Officer	COO																	
NP1FIN	Finance Head	Finance																	
NP1ACC	Accounting	Accounting01																	
NP1PROC	Procurement	Procurement01																	
NP1CHE	CHE Engineer	CHE Engineer 01																	
3	Choose one person group, to show the details of the person group  Will show display like below picture.																		

	 3.a Person Group tab, to view and update details of person group 3.b Detail information of the person group and all person listed in the person group.
4	User can perform some action related to the person from Select Action menu (on top of the form).  Duplicate Person group action to create new person group based on opened person group.

1.3.1 Add/Remove Person in a Person Group

Person group's member modification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to add/remove person groups:

Step	Action
1. Go to Person Groups application by click  (GoTo) – Administration – Resources – Person Groups.	
2. Select one of Person Group to be added or removed	

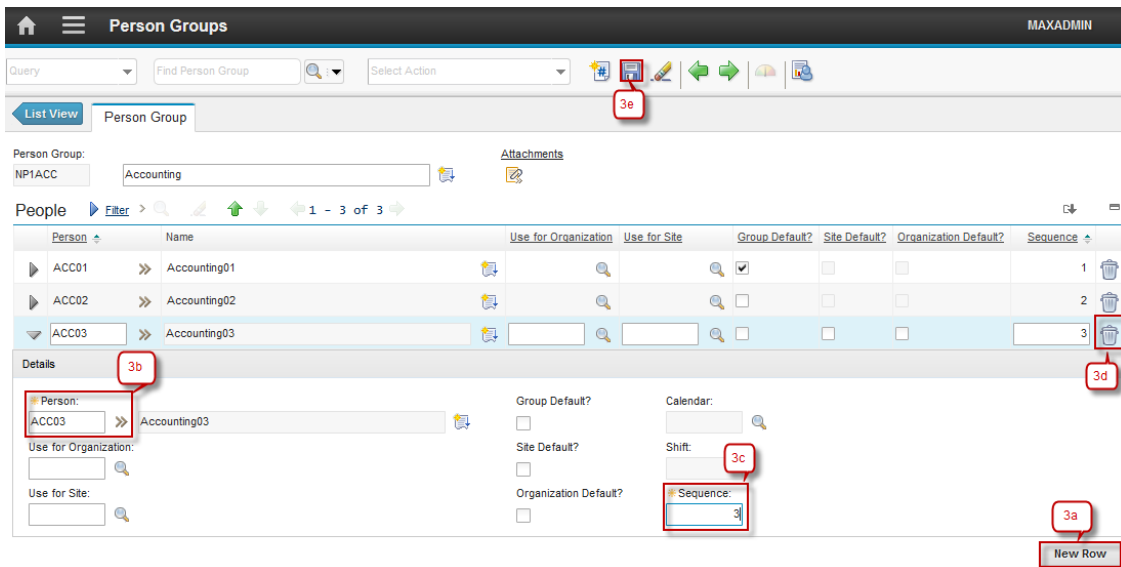
3a. Click **New Row** to add person as member of this person group.

3b. Select the person as member of this person group.

3c. Input the sequence.

3d. Click button  to remove a member of this person group

3e. Click **Save**.



The screenshot shows the 'Person Groups' application interface. At the top, there's a header with a home icon, a menu icon, the title 'Person Groups', and the user 'MAXADMIN'. Below the header is a toolbar with various icons. A 'List View' button is visible. The main content area shows a 'Person Group' section with 'NPIACC' and 'Accounting'. Below this is a 'People' section with a table of members. The table has columns for 'Person', 'Name', 'Use for Organization', 'Use for Site', 'Group Default?', 'Site Default?', 'Organization Default?', and 'Sequence'. Three rows are visible: 'ACC01' (Accounting01), 'ACC02' (Accounting02), and 'ACC03' (Accounting03). The 'ACC03' row is highlighted. To the right of the table is a 'Details' section for 'Accounting03'. It contains fields for 'Person' (set to 'ACC03'), 'Use for Organization', 'Use for Site', 'Group Default?', 'Site Default?', 'Organization Default?', 'Calendar', 'Shift', and 'Sequence'. The 'Sequence' field is highlighted. At the bottom right, there is a 'New Row' button. Red callouts are placed over the interface: '3a' points to the 'New Row' button, '3b' points to the 'Person' dropdown in the details section, '3c' points to the 'Sequence' input field, '3d' points to the trash icon in the table, and '3e' points to the 'Save' button in the toolbar.

1.4 Crafts Application

Crafts application is used to define craft records for a work plan, and to define skill levels, standard rates. The craft code reflects the type of work that employees and contractors perform.

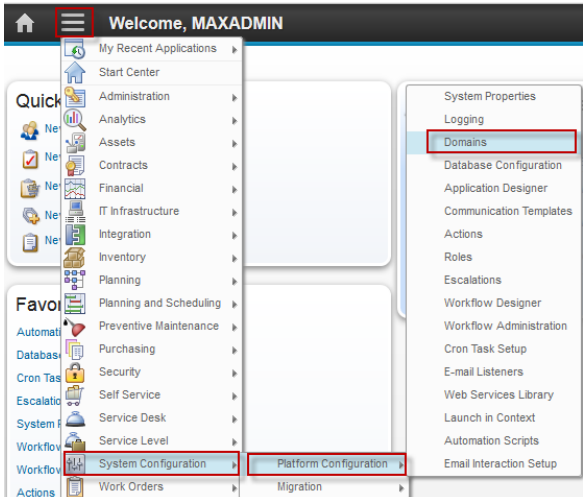
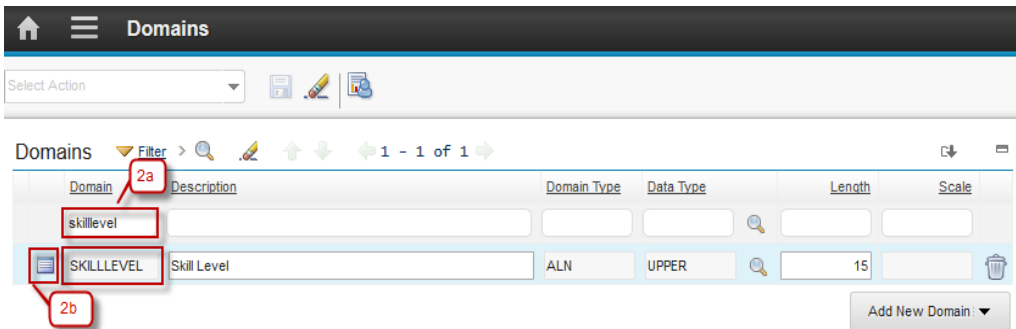
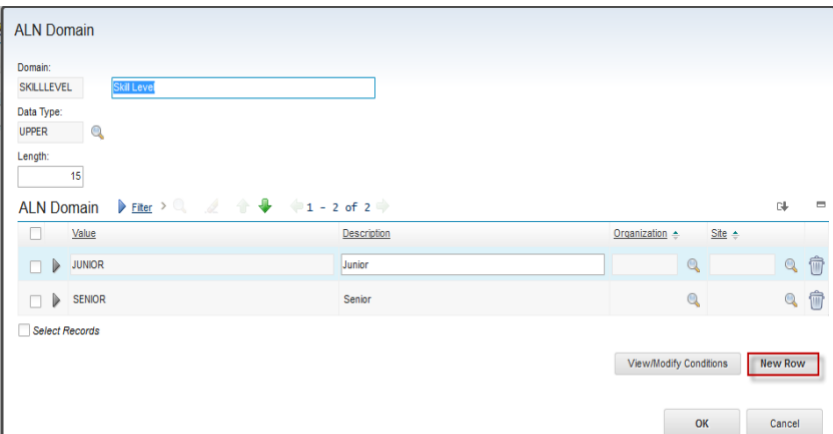
Besides, in **Crafts** application user also can:

- Associate labor records with crafts and skill levels
- Specify multiple skill levels for each craft

To distinguish between a junior level mechanic and a senior level mechanic, it is not required to create two separate craft records. Only one mechanic craft record is needed and set up junior and senior skill levels within the mechanic craft (skill level 1 is highest, and 5 is lowest). Each skill level within the mechanic craft can have different standard rates.

1.4.1 Define Skill Level

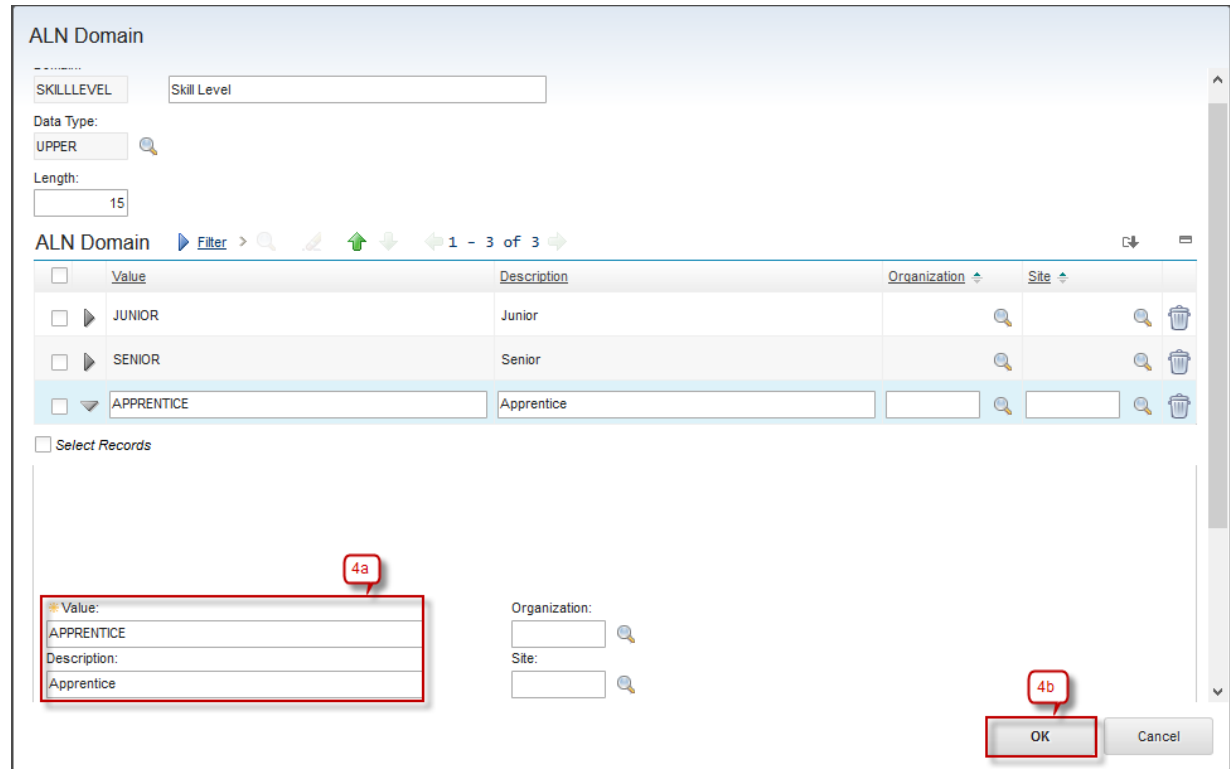
Skill level definition can only be done by *Administrator*. Below are step by step to define Skill Level:

Step	Action
1. Go to Domains application by click  (GoTo) – System Configuration – Platform Configuration – Domains.	
2a. Search Domains name in field Domains = 'skilllevel' 2b. Click button  to add skill level.	
3. Click button New Row	

4a. Input the value of Skill Level.

4b. Input the value description.

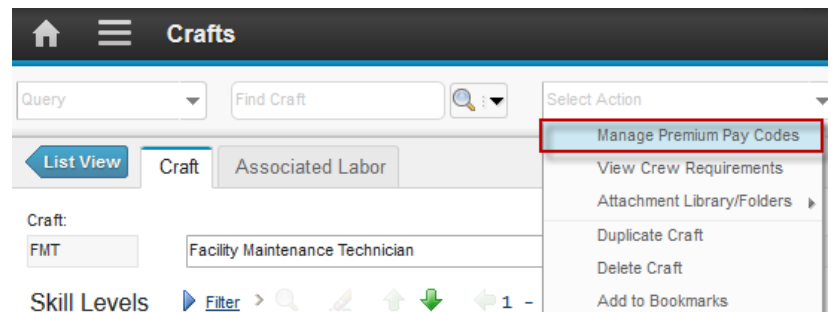
4c. Click button **OK**.



The screenshot shows the 'ALN Domain' form. At the top, there is a 'SKILLLEVEL' field with the value 'Skill Level'. Below it, 'Data Type' is set to 'UPPER' and 'Length' is '15'. A table lists three records: JUNIOR, SENIOR, and APPRENTICE. The 'APPRENTICE' record is selected. Below the table, there is a 'Select Records' checkbox. A red box labeled '4a' highlights the 'Value' field in the 'APPRENTICE' record, which contains 'APPRENTICE'. Another red box labeled '4b' highlights the 'OK' button at the bottom right of the form.

1.4.2 Define Premium Pay Codes

Managing premium pay codes can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to define Premium Pay Codes:

Step	Action
1. Click Craft Tab → Select Action → Manage Premium Pay Codes	 The screenshot shows the 'Crafts' tab interface. The 'Select Action' dropdown menu is open, and the 'Manage Premium Pay Codes' option is highlighted with a red box. Other options in the menu include 'View Crew Requirements', 'Attachment Library/Folders', 'Duplicate Craft', 'Delete Craft', and 'Add to Bookmarks'.

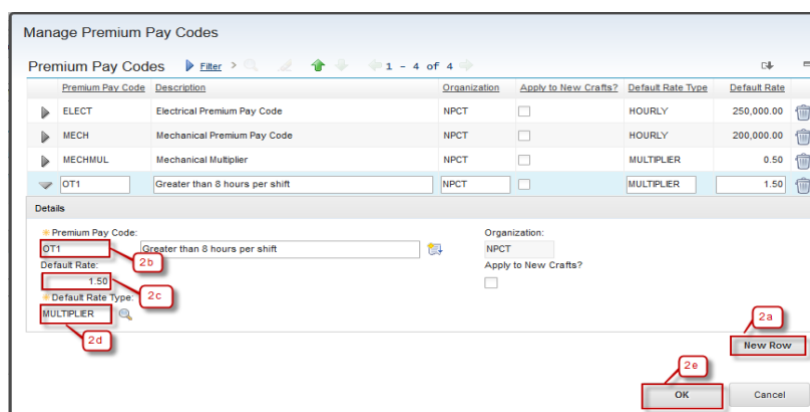
2a. Click **New Row**.

2b. Input the Premium Pay Code.

2c. Input Default Rate.

2d. Input Default Rate Type.

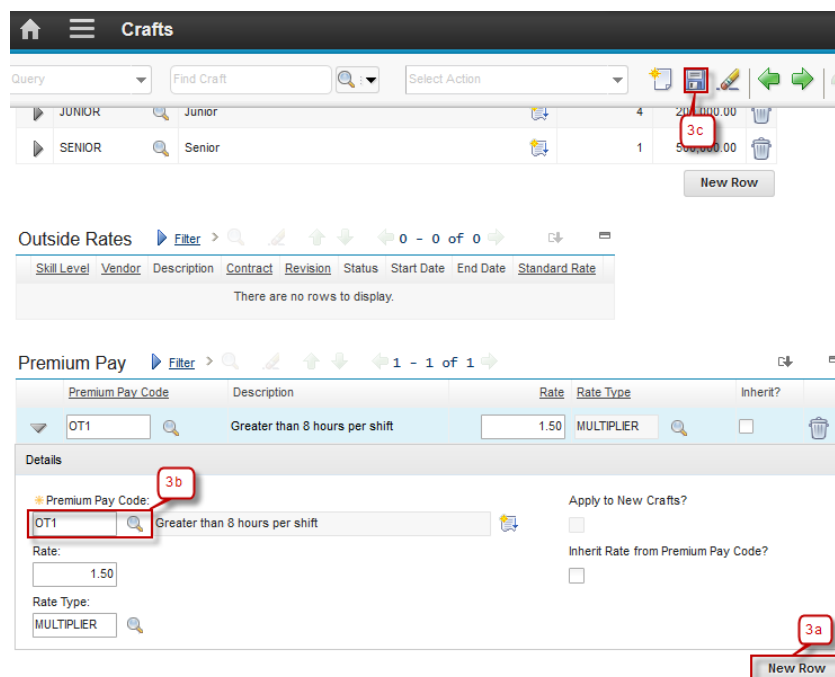
2e. Click **OK**.



3a. Click **New Row** to add Premium Pay in this Craft.

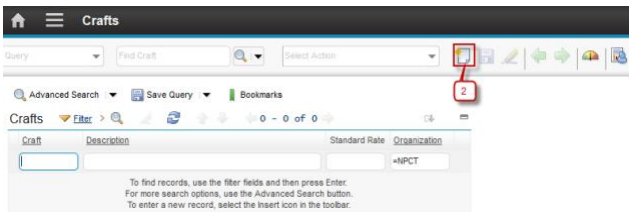
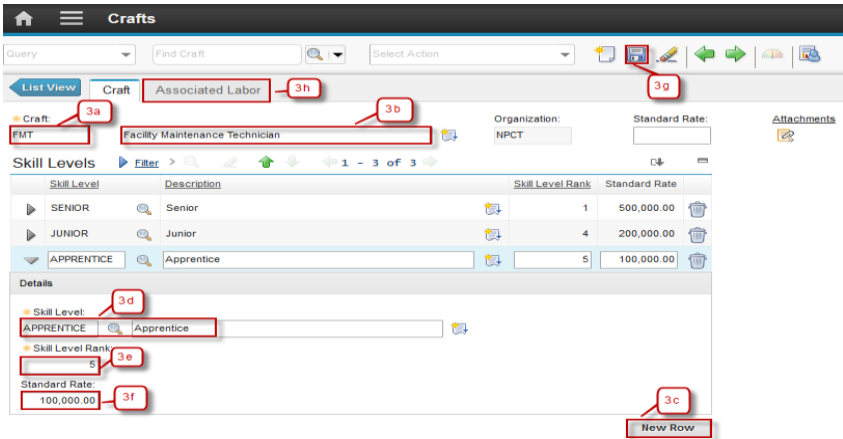
3b. Select Premium Pay Code using button 

3c. Click **Save**

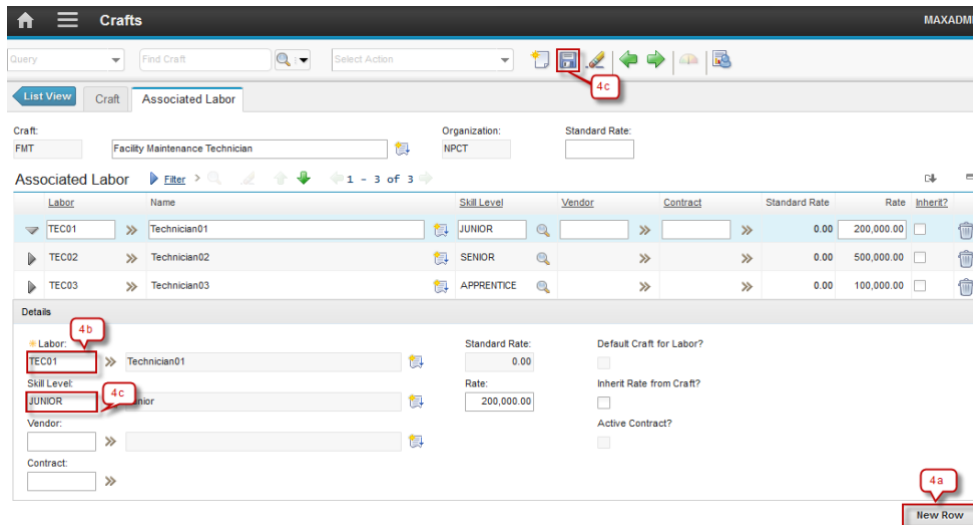


1.4.3 Create Crafts

Craft creation can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Crafts:

Step	Action
1. Go to Crafts application by click  (GoTo) – Administration – Resources – Crafts.	
2. Click  New Craft .	
3a. Input the Craft name. 3b. Input the Craft description. 3c. Click New Row to add Skill Level of this craft. 3d. Input the Skill Level name and the description. 3e. Input the Skill Level Rank (1 is highest, 5 is lowest). 3f. Input Standard Rate for that skill level of that craft. 3g. Click Save. 3h. Click Associated Labor tab to input labor that has this craft skill.	

- 4a. Click **New Row** to associate labor with this craft skill.
- 4b. Input the labor name that has this craft skill.
- 4c. Input the skill level of this craft that owned by the labor.
- 4d. Click **Save**.



The screenshot shows the 'Crafts' application interface. At the top, there's a header with 'Crafts' and a user name 'MAXADMIN'. Below the header is a search bar and a 'Select Action' dropdown. The main section is titled 'Associated Labor' and contains a table with the following data:

Labor	Name	Skill Level	Vendor	Contract	Standard Rate	Rate	Inherit?
TEC01	Technician01	JUNIOR			0.00	200,000.00	☐
TEC02	Technician02	SENIOR			0.00	500,000.00	☐
TEC03	Technician03	APPRENTICE			0.00	100,000.00	☐

Below the table is a 'Details' section with the following fields:

- Labor: TEC01
- Skill Level: JUNIOR
- Vendor:
- Contract:
- Standard Rate: 0.00
- Rate: 200,000.00
- Default Craft for Labor? ☐
- Inherit Rate from Craft? ☐
- Active Contract? ☐

Red callouts indicate the steps: 4a points to the 'New Row' button, 4b points to the 'Labor' field, 4c points to the 'Skill Level' field, and 4d points to the 'Save' button.

1.5 Labor Application

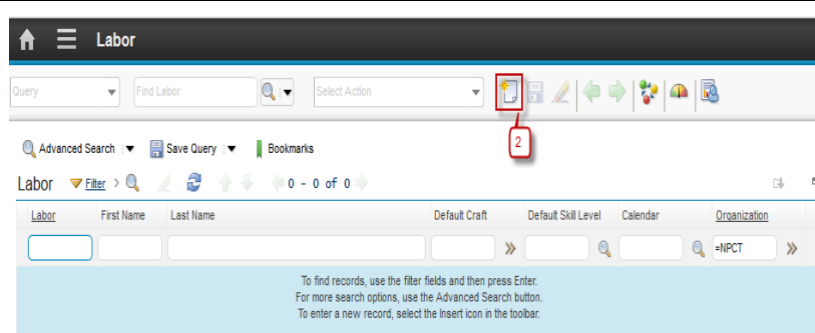
Labor application is used to create, change, view, and delete labor records for employees and contractors. Labor records can contain personal and work-related information, as well as the type and location of work. In addition, crafts, skill levels, qualifications, and certifications can be specified that are associated with labor records. Before labor record can be created, a person record corresponding to the labor must be created first.

1.5.1 Create Labors

Labor data creation can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Labor:

Step	Action
1. Go to Labor application by click  (GoTo) – Administration – Resources – Labor .	

2. Click  **New Labor.**



The screenshot shows the 'Labor' management interface. At the top, there's a header with a home icon, a menu icon, and the word 'Labor'. Below this is a toolbar with various icons. The 'New Labor' icon, which is a document with a plus sign, is highlighted with a red box and labeled with the number '2'. Below the toolbar, there are search and filter options, and a table with columns for Labor, First Name, Last Name, Default Craft, Default Skill Level, Calendar, and Organization. The table is currently empty, and a message at the bottom suggests using filter fields or the Advanced Search button to find records.

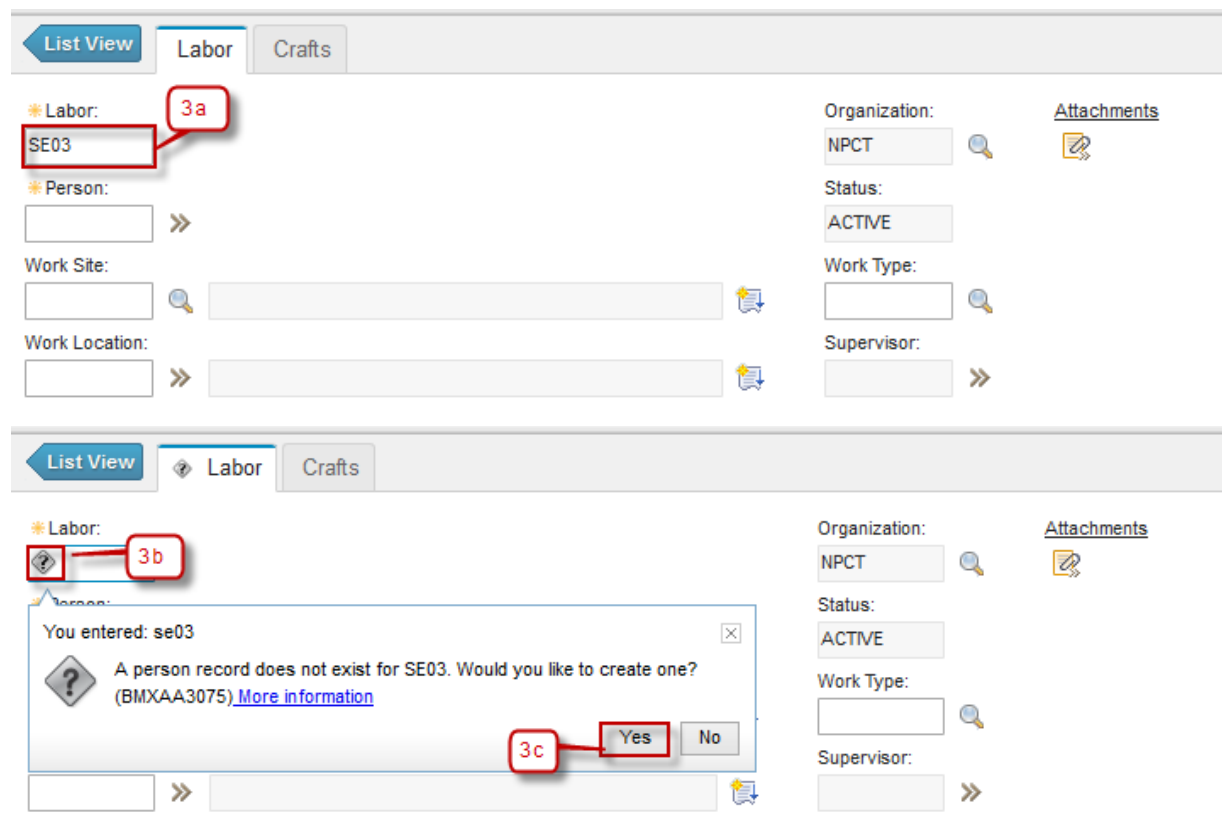
3a. Input the Labor Code.

3b. If the labor code that inputted above doesn't have a person data yet, the question mark icon will appear, Click the question mark.

3c. Click **Yes** to create the person data based on the labor code that inputted above

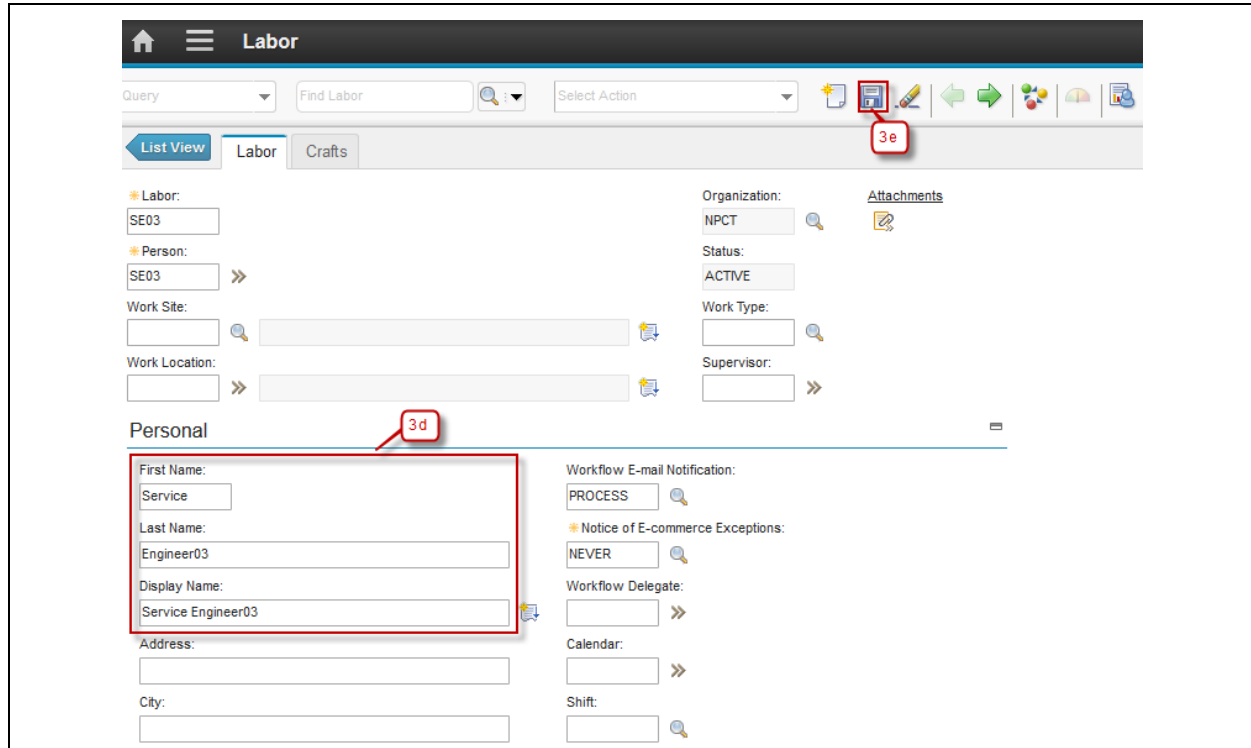
3d. Input any additional information about Labor

3e. Click **Save.**



The first screenshot shows the 'New Labor' form. The 'Labor' field is highlighted with a red box and labeled '3a', containing the value 'SE03'. The 'Person' field is empty. The 'Work Site' and 'Work Location' fields are also empty. The 'Organization' field is set to 'NPCT', and the 'Status' is 'ACTIVE'. The 'Work Type' and 'Supervisor' fields are empty. The 'Attachments' link is visible on the right.

The second screenshot shows the same form, but the 'Labor' field now has a question mark icon next to it, highlighted with a red box and labeled '3b'. A dialog box is open, asking 'A person record does not exist for SE03. Would you like to create one? (BMXAA3075) More information'. The 'Yes' button in the dialog box is highlighted with a red box and labeled '3c'.



Home Menu Labor

Query Find Labor Select Action

List View Labor Crafts

Labor: SE03

Person: SE03 >>

Work Site: >>

Work Location: >>

Organization: NPCT

Status: ACTIVE

Work Type: >>

Supervisor: >>

Attachments

Personal

First Name: Service

Last Name: Engineer03

Display Name: Service Engineer03

Address: >>

City: >>

Workflow E-mail Notification: PROCESS

Notice of E-commerce Exceptions: NEVER

Workflow Delegate: >>

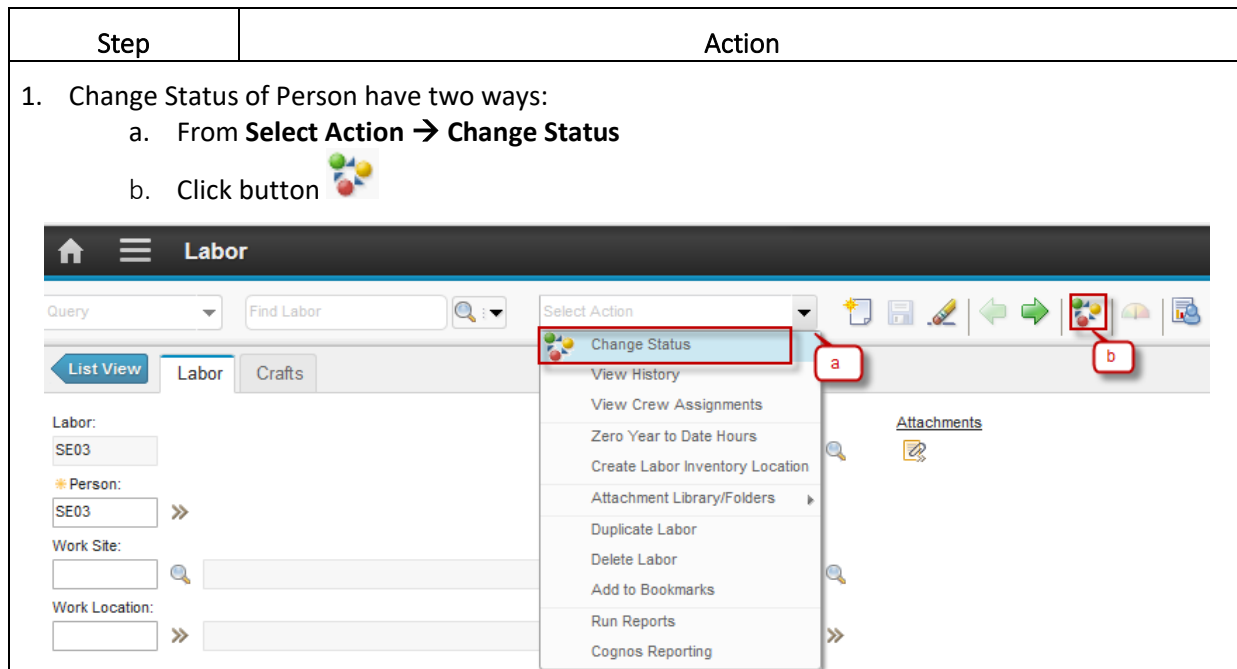
Calendar: >>

Shift: >>

1.5.2 Change Status of Labor

Labor status change can be done by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Change Status of Person

Step	Action
1. Change Status of Person have two ways:	
a. From Select Action → Change Status	
b. Click button 	



Home Menu Labor

Query Find Labor Select Action

List View Labor Crafts

Labor: SE03

Person: SE03 >>

Work Site: >>

Work Location: >>

Organization: NPCT

Status: ACTIVE

Work Type: >>

Supervisor: >>

Attachments

Change Status

View History

View Crew Assignments

Zero Year to Date Hours

Create Labor Inventory Location

Attachment Library/Folders

Duplicate Labor

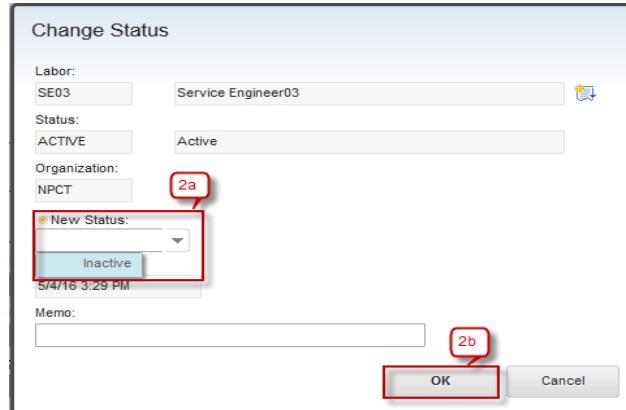
Delete Labor

Add to Bookmarks

Run Reports

Cognos Reporting

2a. Dialog box **Change Status** will be appeared, see the figure below :

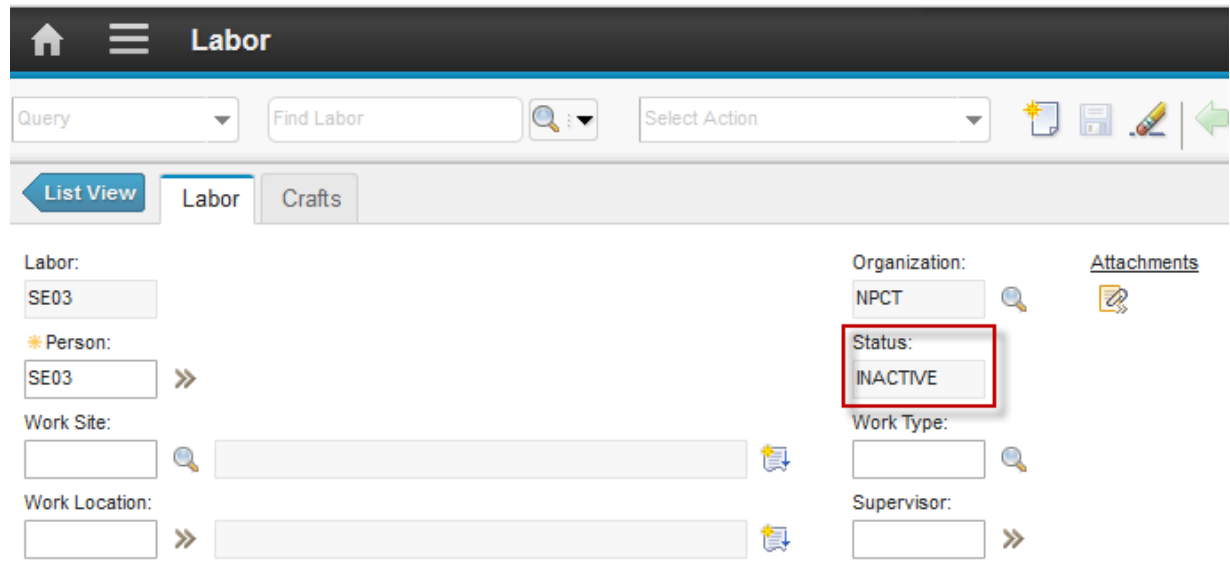


The 'Change Status' dialog box contains the following fields and controls:

- Labor:** SE03 (text field), Service Engineer03 (text field)
- Status:** ACTIVE (text field), Active (text field)
- Organization:** NPCT (text field)
- New Status:** A dropdown menu with 'Inactive' selected. A red box labeled '2a' highlights this dropdown.
- Memo:** A text area.
- Buttons:** OK and Cancel. A red box labeled '2b' highlights the OK button.

2b. Click button **OK**

4. Status will be changed to **INACTIVE**



The 'Labor' management interface shows the following details:

- Header:** Labor
- Query:** Find Labor (text field), Select Action (dropdown menu)
- Views:** List View, Labor, Crafts (tabs)
- Labor Details:**
 - Labor:** SE03
 - Person:** SE03 (with a double arrow icon)
 - Work Site:** (text field)
 - Work Location:** (text field)
- Organization:** NPCT
- Status:** INACTIVE (highlighted with a red box)
- Work Type:** (text field)
- Supervisor:** (text field)
- Attachments:** (link)

1.6 Exercise Phase

2 Asset & Location

2.1 Overview

Asset is an item of economic value that a company own, benefit from, or has use of, in generating income. In EMS, an item is treated as asset when user wants to track its moving history, and maintenance work can occurs toward it.

Asset can be either rotating or non-rotating. Rotating assets are assets that are interchangeable, such as motors, pumps, or fire extinguishers. In EMS, remarkable difference between rotating asset and non-rotating asset is that rotating assets have both an asset number and an inventory item number, while non-rotating just have an asset number, without an inventory item number. The item number lets user to track assets as a group as they are moved in and out of inventory and other types of locations, while the asset number is useful to track individual instances of asset as it is moved from one location to another and from one site to another.

Here is example to help understanding inventory item number and asset. A company might have four identical (same make, same model) centrifugal pumps, so all four pumps would have the same item number. To track the use, repair, and locations of each individual pump, each pump would have its own, unique asset number.

Location is generally defined as a place where assets are operated, stored, and repaired. Generally locations are defined as a means of tracking assets. Another consideration, a maintenance work can be done toward **Locations**.

2.2 Classification Application

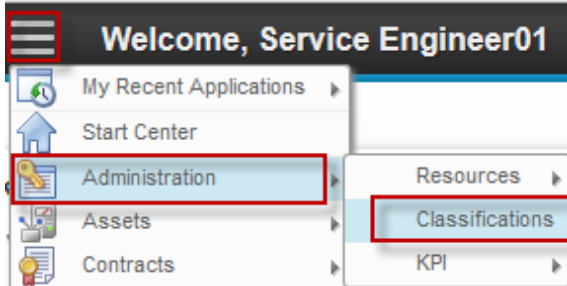
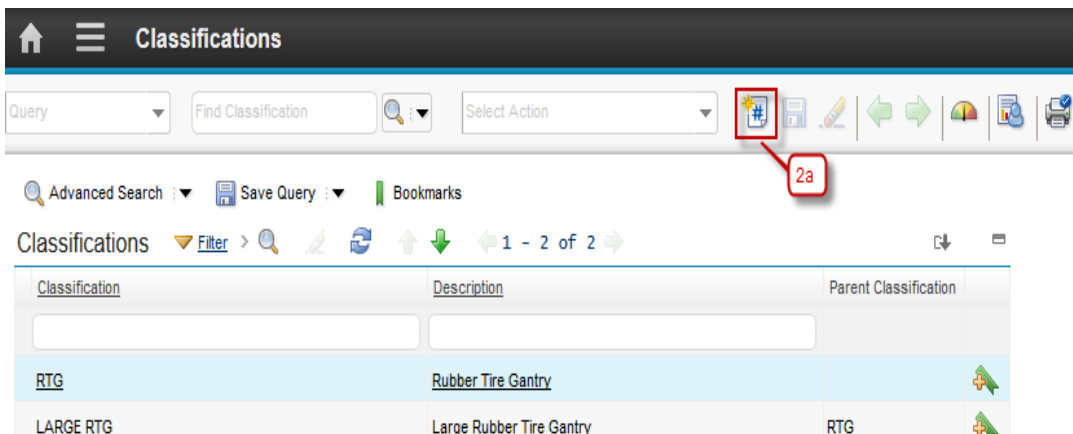
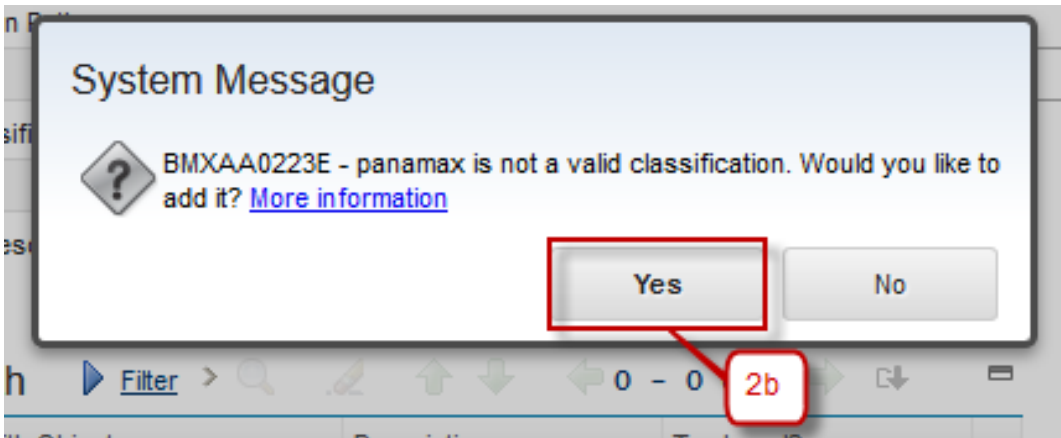
The Specifications tab on asset and location records can be used to add Classification information to asset and location records. Specification is available on item data too. Using Classifications lets user to structure assets and locations in organized hierarchies so that:

- Records can be easily located via Classifications.
- Records are not duplicated unintentionally.
- Asset and item lists are consistent with vendor's lists.

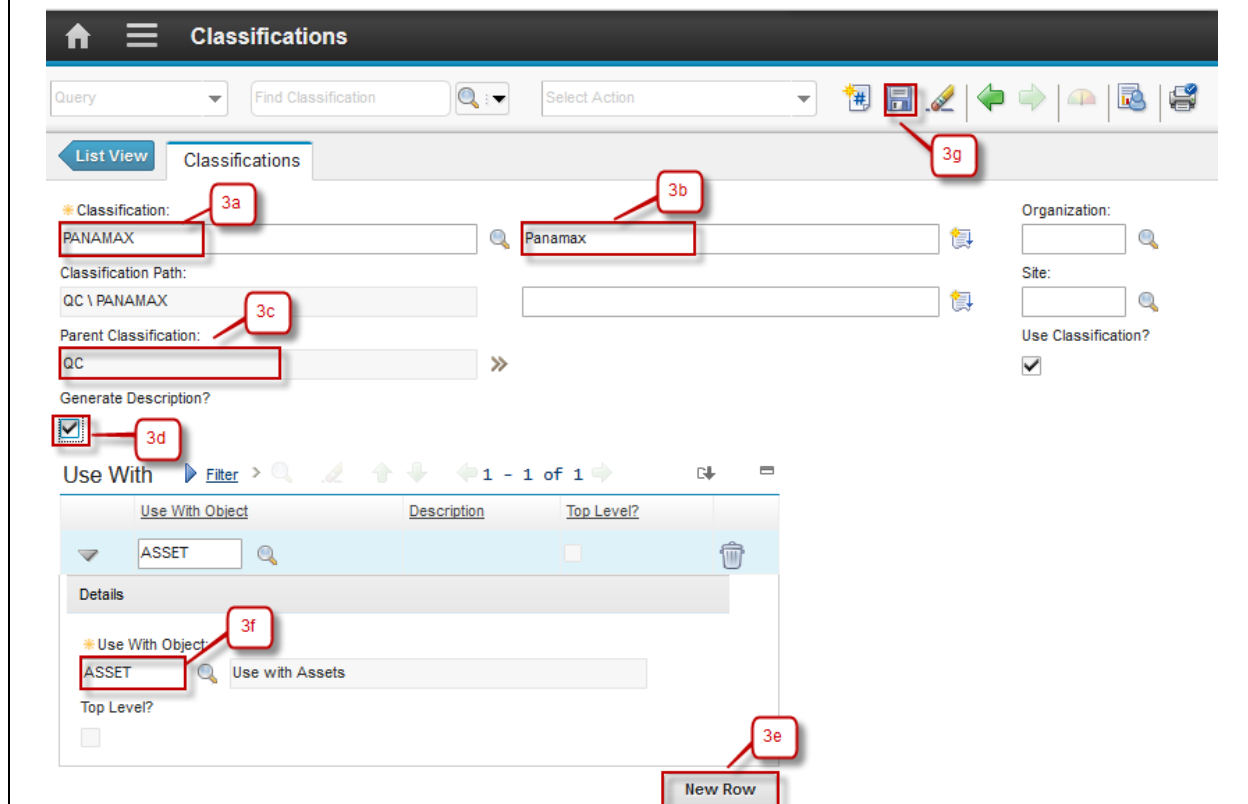
Classifications use attributes to define the classification category. When you list a rotating item that has a classification on an asset or location record, Maximo copies the Classification and Attributes to **Specifications** tab of the asset or location record.

2.2.1 Create Classifications

Classification creation can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Classification:

Step	Action
1. Go to Classification application by click  (GoTo) – Administration – Classifications.	
2a. Click New Classification .	 
2b. If the classification that inputted above doesn't valid, the message box to added the classification will appear. Click button Yes and then classification will be added.	

- 3a. Input the Classification name.
- 3b. Input the Classification description.
- 3c. Input the Parent Classification if this classification has a parent.
- 3d. Default from **Generate Description** is checked, if description classification don't want generated when the classification in use make sure the check box is unchecked.
- 3e. Click **New Row** to add the object that will use this classification.
- 3f. Input the object that will use this classification.
- 3g. Click **Save**.



The screenshot shows the 'Classifications' management interface. At the top, there's a header with a home icon, a menu icon, and the title 'Classifications'. Below this is a toolbar with a 'Query' dropdown, a 'Find Classification' search bar, a 'Select Action' dropdown, and several icons including a plus sign (labeled 3g), a save icon, a delete icon, and navigation arrows. The main area is divided into a 'List View' tab and a 'Classifications' tab. In the 'Classifications' tab, there are several input fields: 'Classification:' with the value 'PANAMAX' (labeled 3a), 'Classification Path:' with the value 'QC \ PANAMAX' (labeled 3c), and 'Parent Classification:' with the value 'QC' (labeled 3c). To the right of these fields are 'Organization:', 'Site:', and 'Use Classification?' (checked) fields. Below the input fields is a 'Generate Description?' checkbox (checked, labeled 3d). At the bottom, there is a 'Use With' section with a table showing 'Use With Object' (ASSET), 'Description' (Use with Assets), and 'Top Level?' (unchecked). A 'New Row' button is at the bottom right (labeled 3e). A 'Details' section is also visible below the table, showing 'Use With Object' (ASSET, labeled 3f) and 'Top Level?' (unchecked).

2.2.2 Define Attributes to a Classification

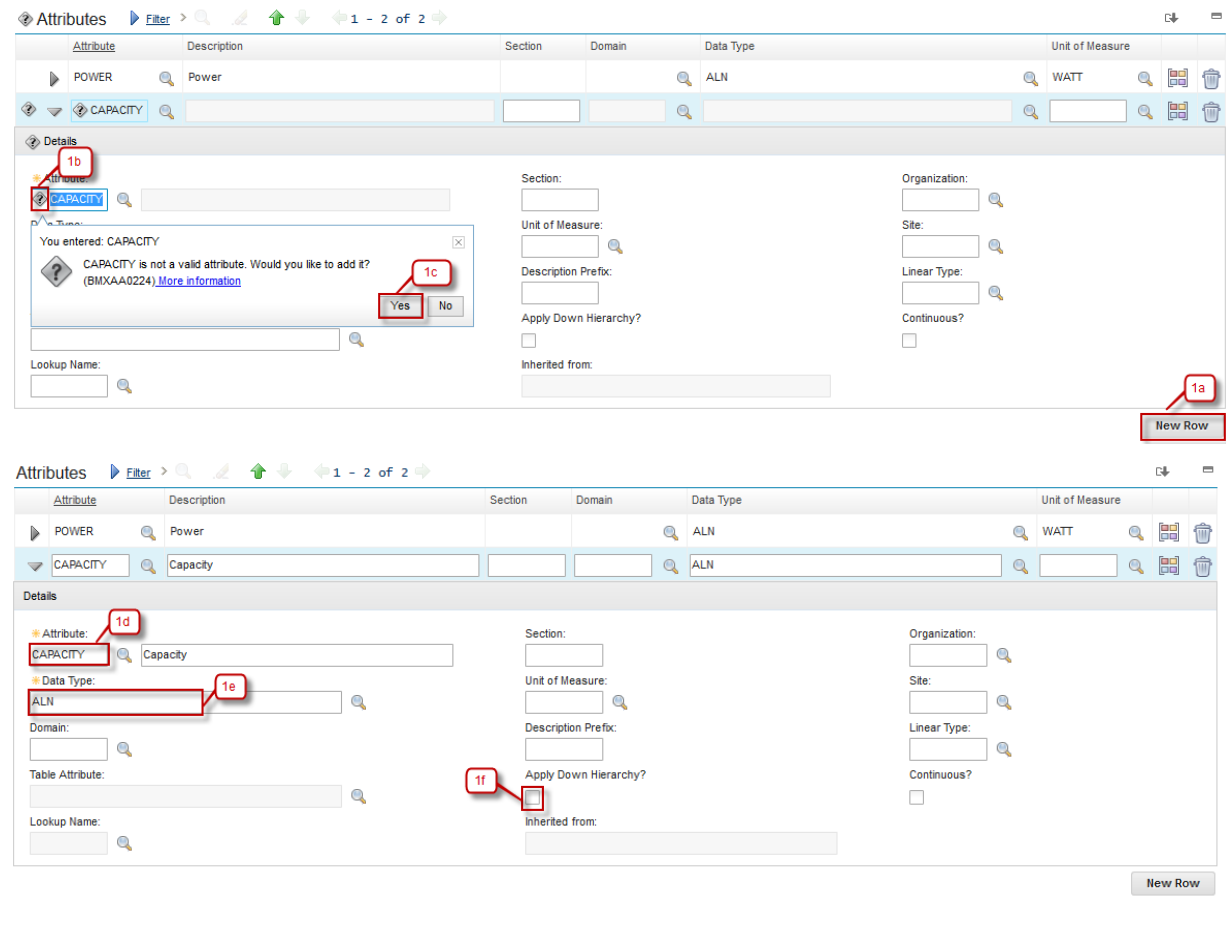
Defining attributes to a classification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Define Attribute to a Classification:

Step	Action
1a. Click New Row to add attribute on this classification.	
1b. If the attribute you added is a new attribute, question mark icon will appear.	
1c. Click Yes to add the attribute.	

1d. Input the Description of that attribute.

1e. Input the Data Type of that attribute.

1f. Check the **Apply Down Hierarchy** checkbox if you want this attribute will appear in any child classification of this classification and click **Save**.



The screenshot shows the 'Attributes' application interface. At the top, there's a table with columns: Attribute, Description, Section, Domain, Data Type, and Unit of Measure. The first row is 'POWER' with description 'Power', domain 'ALN', and unit 'WATT'. The second row is 'CAPACITY' with description 'Capacity', domain 'ALN', and unit 'WATT'. Below the table is a 'Details' form for adding a new attribute. Red callouts highlight the following steps:

- 1a**: Clicking the 'New Row' button at the bottom right of the table.
- 1b**: Inputting 'CAPACITY' into the 'Attribute' field.
- 1c**: Clicking 'Yes' in the dialog box that says 'CAPACITY is not a valid attribute. Would you like to add it? (BMXAA0224) More information'.
- 1d**: Inputting 'Capacity' into the 'Description' field.
- 1e**: Inputting 'ALN' into the 'Data Type' field.
- 1f**: Checking the 'Apply Down Hierarchy' checkbox.

2.3 Locations Application

Locations application is used to add, view, modify, and delete location records for assets, and organize these location into systems. Systems are used to maintain location's hierarchical. When location is organized into systems, user can quickly find a location using the drilldown and then identify the assets at a specific location.

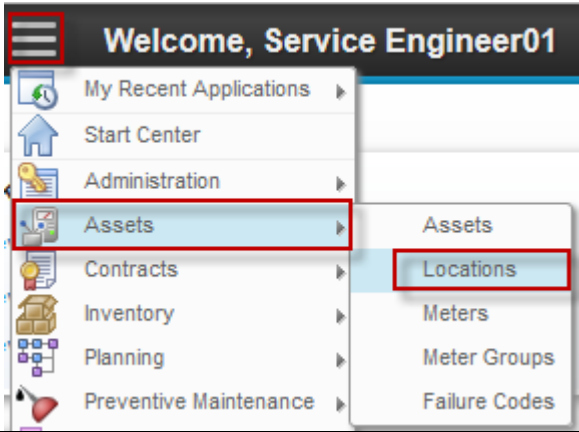
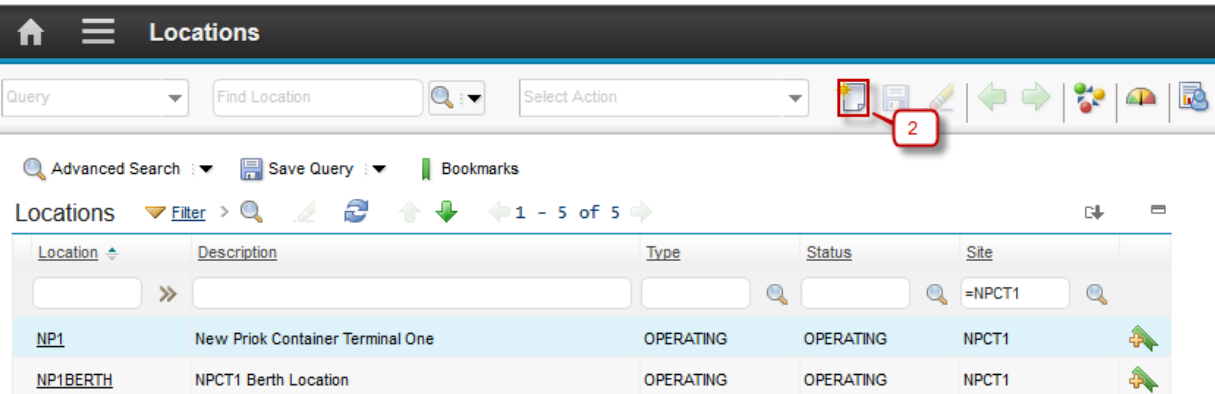
Locations are categorized into several types:

- **Courier** - This is an inventory location, used to indicate an individual who is responsible for inventory items while in transit from one inventory location to another.

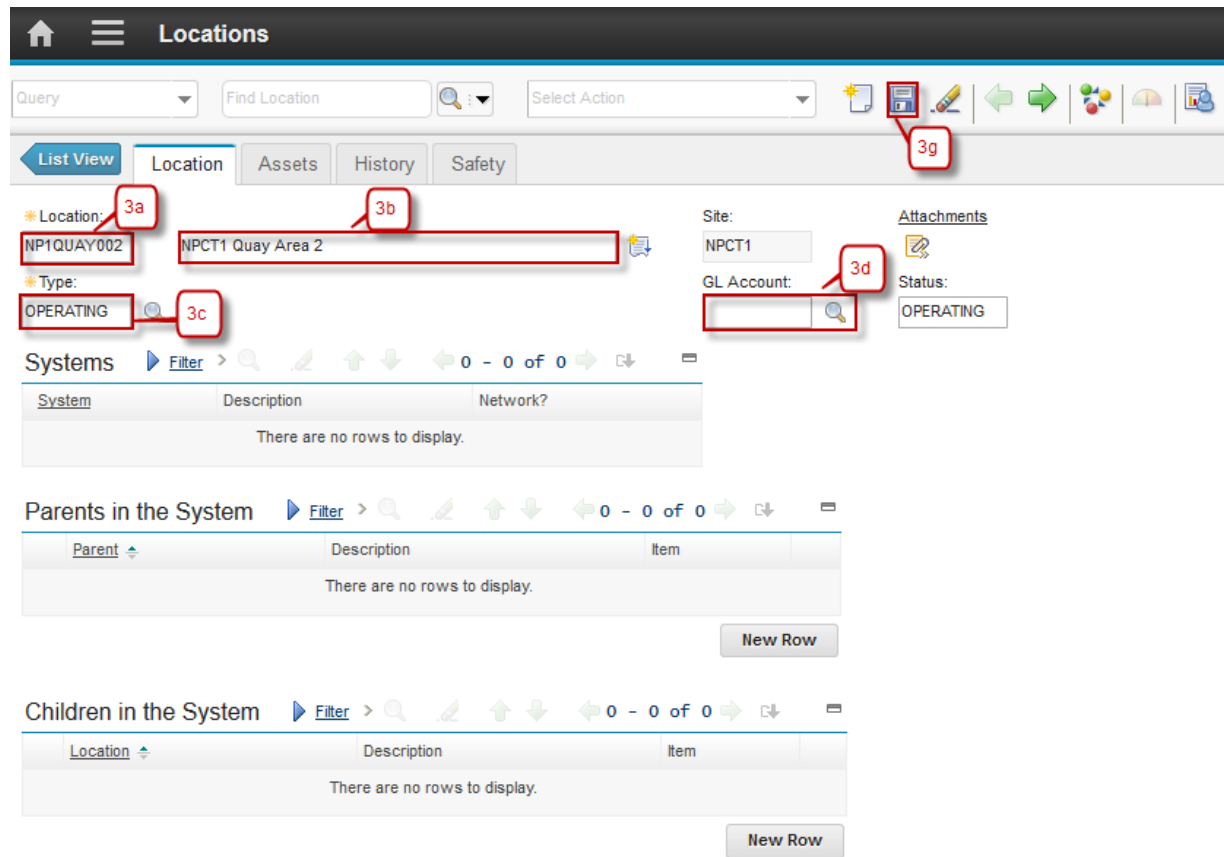
- Holding - This is an inventory location, used to hold items received with a status of WINSP (waiting for inspection). One site can only have one holding location.
- Operating - This is an asset location, used to indicate a location where assets are located or equipment is operated.
- Repair - This is an asset location, used to indicate a location where assets are repaired or refurbished.
- Salvage - This is an asset location, used to indicate a location used to store retired or decommissioned assets.
- Vendor - This is an inventory location, used to indicate the source of items, material, and rotating assets, and may be used to indicate the location of an asset returned to the vendor for repair or refurbishment.

2.3.1 Create Locations

Location creation can be performed by *Service Engineer*. Below are step by step to Create Locations:

Step	Action
1. Go to Locations application by click (GoTo) – Assets – Locations.	
2. Click  New Locations .	

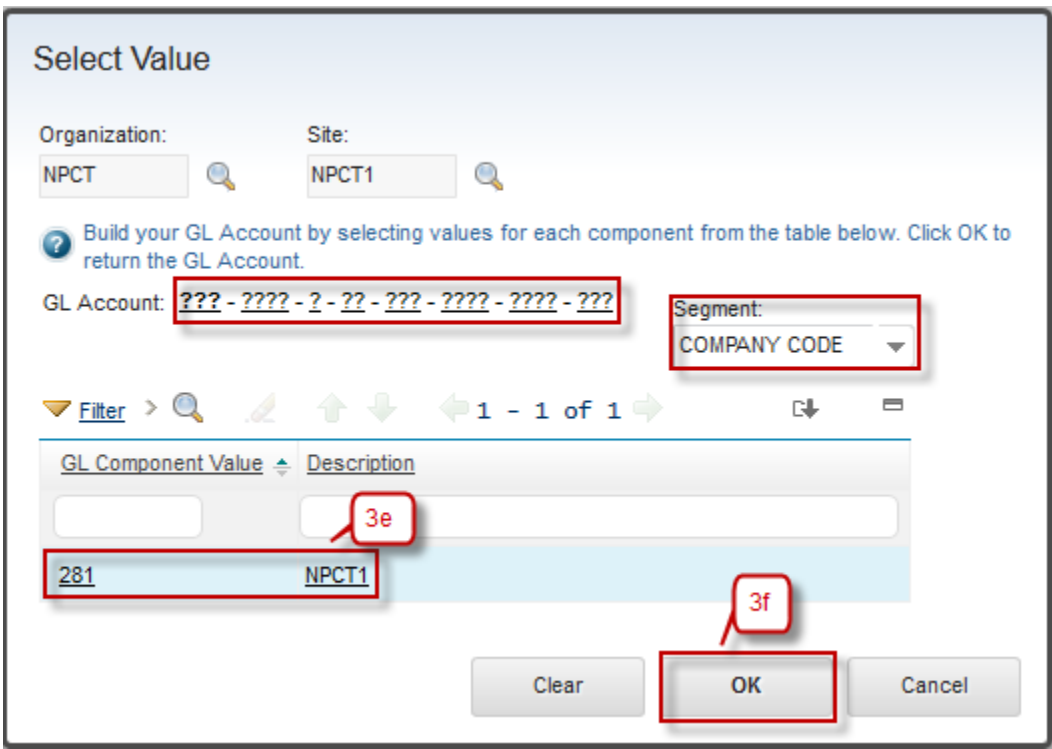
- 3a. Input the Locations name.
- 3b. Input the Locations description.
- 3c. Select the type of Locations, default type is 'OPERATING'.
- 3d. Define GL Account of Locations, click button 
- 3e. Select account value for every segment of GL Account.
- 3f. Click **OK**.
- 3g. Click **Save**.



The screenshot shows the 'Locations' form in the Agile system. The form is titled 'Locations' and has a navigation bar with tabs: 'List View', 'Location', 'Assets', 'History', and 'Safety'. The 'Location' tab is active. The form contains several input fields and buttons:

- Location:** A text field containing 'NP1QUAY002' (annotated with 3a).
- Description:** A text field containing 'NPCT1 Quay Area 2' (annotated with 3b).
- Type:** A dropdown menu with 'OPERATING' selected (annotated with 3c).
- Site:** A text field containing 'NPCT1'.
- GL Account:** A text field with a magnifying glass icon (annotated with 3d).
- Status:** A dropdown menu with 'OPERATING' selected.
- Attachments:** A button with a plus icon.
- Buttons:** 'Save' (annotated with 3g), 'OK' (3f), and 'Cancel'.

Below the form, there are three sections: 'Systems', 'Parents in the System', and 'Children in the System'. Each section has a table with columns for 'System', 'Description', and 'Network?'. All three sections show 'There are no rows to display.' and a 'New Row' button.



Select Value

Organization: NPCT Site: NPCT

Build your GL Account by selecting values for each component from the table below. Click OK to return the GL Account.

GL Account: ??? - ??? - ? - ?? - ??? - ??? - ??? Segment: COMPANY CODE

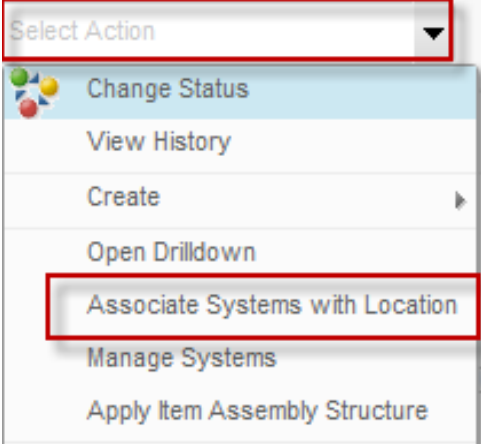
Filter > 1 - 1 of 1

GL Component Value	Description
281	NPCT1

Clear OK Cancel

2.3.2 Associate Systems with Locations

Below are step by step to Associate Systems with Locations by *Service Engineer*:

Step	Action
1. Go to Select Action – Associate Systems with Location.	

- 2a. Click **New Row**.
- 2b. Select System of Locations.
- 2c. Select Parent of Locations.
- 2d. Click button **OK**.

Associate Systems with Location

To associate a new system with the current location, click New Row.

Location: NP1QUAY002 NPCT1 Quay Area 2

Site: NPCT1

System Filter > 1 - 1 of 1

System	Description	Primary System?	Network?	Parent
PRIMARY	Primary System	☒	☐	NP1QUAY001

Details

* System: PRIMARY Primary System

Parent: NP1QUAY001 NPCT1 Quay Area

Primary System? ☒

Network? ☐

2a New Row

2b

2c

2d OK Cancel

3. Field System and Parent will defined

List View Location Assets History Safety

Location: NP1QUAY002 NPCT1 Quay Area 2

Site: NPCT1

Type: OPERATING

GL Account: 281-0091-0-01

Attachments

Status: OPERATING

Systems Filter > 1 - 1 of 1

System	Description	Network?
PRIMARY	Primary System	☐

Parent of NP1QUAY002 in the PRIMARY System Filter > 1 - 1 of 1

Parent	Description	Item
NP1QUAY001	NPCT1 Quay Area	

Children of NP1QUAY002 in the PRIMARY System Filter > 0 - 0 of 0

Location	Description	Item
There are no rows to display.		

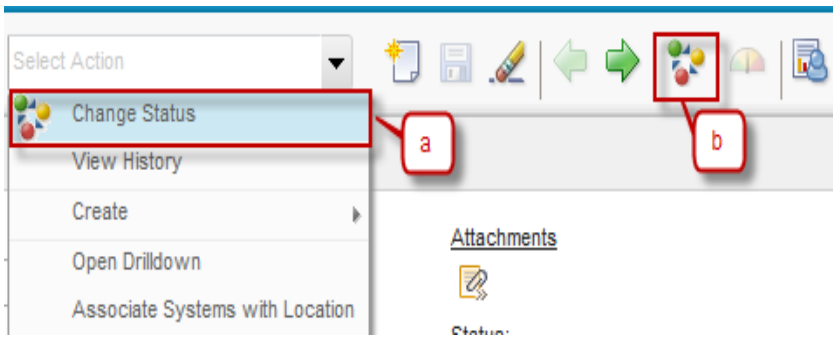
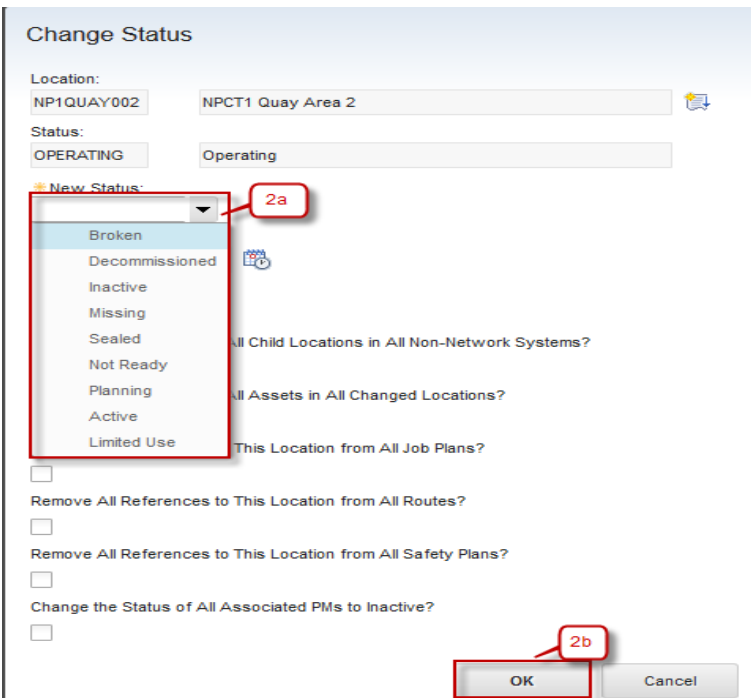
3

New Row

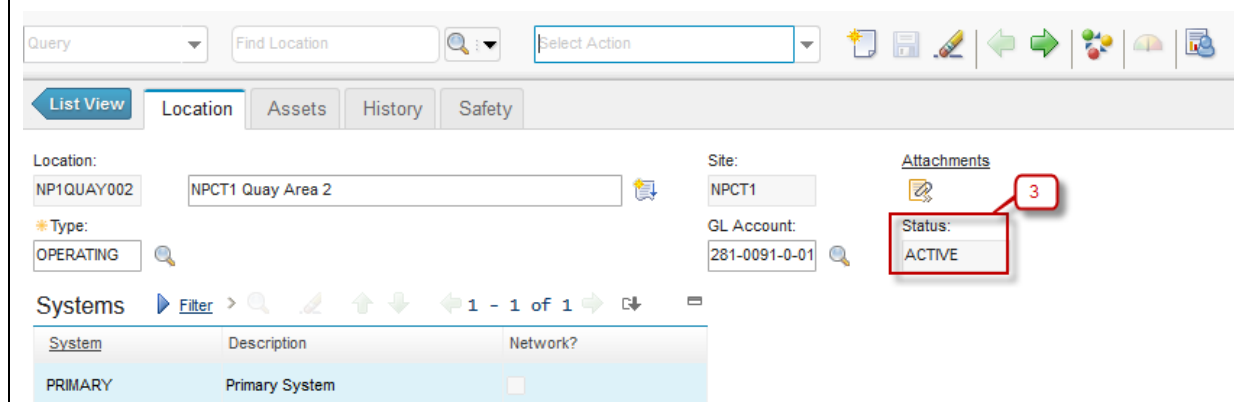
New Row

2.3.3 Change Status of a Location

Below are step by step to Change Status of a Location by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From Select Action → Change Status b. Click button 	
2. Dialog box Change Status will be appeared, see the figure below : 2a. Select New Status of Locations. 2b. Click OK .	

3. Status will be change to **ACTIVE**



The screenshot shows the Agile Solution interface for managing assets. At the top, there are search and action fields. Below, the 'Location' tab is active, showing details for 'NP1QUAY002' and 'NPCT1 Quay Area 2'. The 'Type' is set to 'OPERATING'. The 'Site' is 'NPCT1' and the 'GL Account' is '281-0091-0-01'. On the right, the 'Attachments' section shows a red box around the 'Status' field, which is currently 'ACTIVE'. A red circle with the number '3' is next to it, indicating a step in the process. Below this, the 'Systems' table is visible, showing a single entry for 'PRIMARY' with the description 'Primary System'.

System	Description	Network?
PRIMARY	Primary System	☐

2.4 Meters & Meter Groups Application

A meter record in EMS is virtual identifier used to record measurements. meter records is associated with assets and locations to record measurements, for example run hours, temperature, or pressure, that are used to track asset and location performance. **Meters** application is used to add, view, modify, and delete meter records.

Meters can have one of three types:

- Continuous - Meters where the readings increase continuously, for example odometers. These meters might be used to track miles, hours, engine starts, pieces produced, fuel consumed, and other continuous readings.
- Gauge - Meters where the readings may fluctuate, for example thermometers or pressure gauges. These meters might be used to track temperature, pressure, noise levels, oil levels, and other readings that fluctuate,
- Characteristic - Meters where an observed state is being tracked, for example a color change. These meters might be used to track Brick/Refractory Color (yellow, orange, white), or oil color (clear, turbid, dark).

Meter readings may be recorded for a variety of reasons, including:

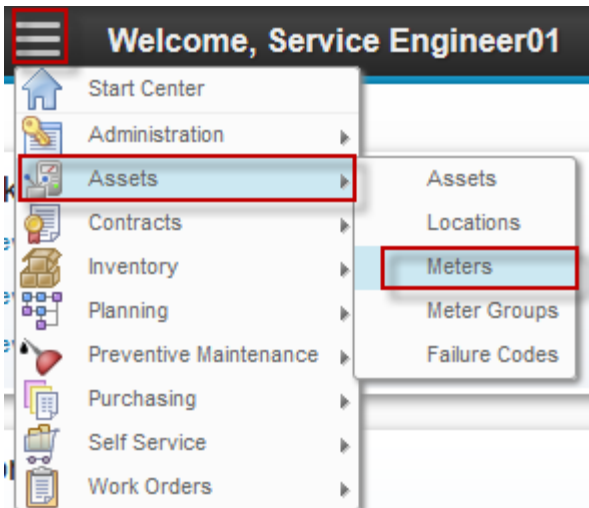
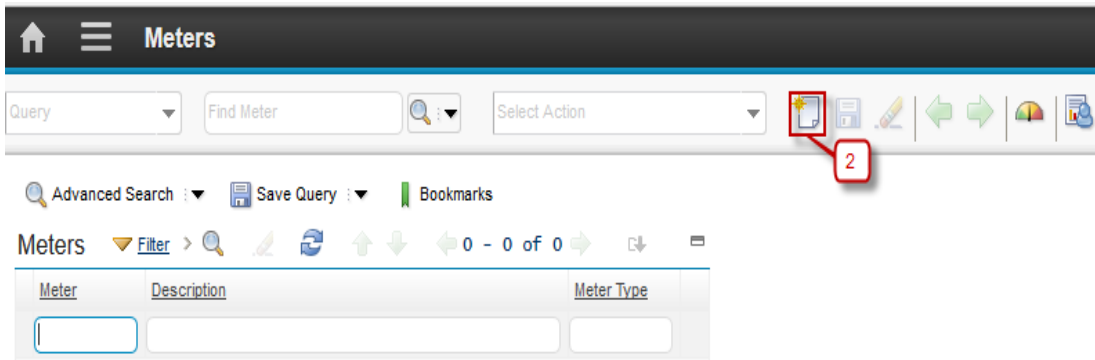
- determining the next time an asset is due for maintenance
- performing efficiency comparisons between similar asset performing similar work
- tracking production costs, for example tons per hour in the mining industry
- tracking service costs, for example odometer readings for a trucking fleet
- tracking efficiency indicators, for example miles per gallon
- conforming to regulatory requirements, for example maintaining cradle to grave details of time controlled components in the aviation industry

In **Meters** application, when user create a meter record, user can define the type of meter, the unit of measure used by the meter, and the type of reading (delta or actual) that the meter should record. Meter records can be associated with assets, inventory items, and locations, and can be used to drive condition monitoring and preventive maintenance. Meters can be applied individually, or as part of a meter group.

A meter group is two or more meters that are commonly used on the same types of assets or locations. **Meter Groups** application is used to create groups for meters that are commonly used together. For example all of the vehicles in a fleet use meters to track mileage, usage, fuel, and oil consumption. Rather than attach these four meters to a vehicle one at a time, user can define a meter group and use it to associate all four meters to the vehicle at the same time.

2.4.1 Create Meters

Below are step by step to Create Meters by *Service Engineer*:

Step	Action
1. Go to Meters application by click  (GoTo) – Assets – Meters.	
2. Click  New Row.	

3a. Input the Meter name.

3b. Input the Meter description.

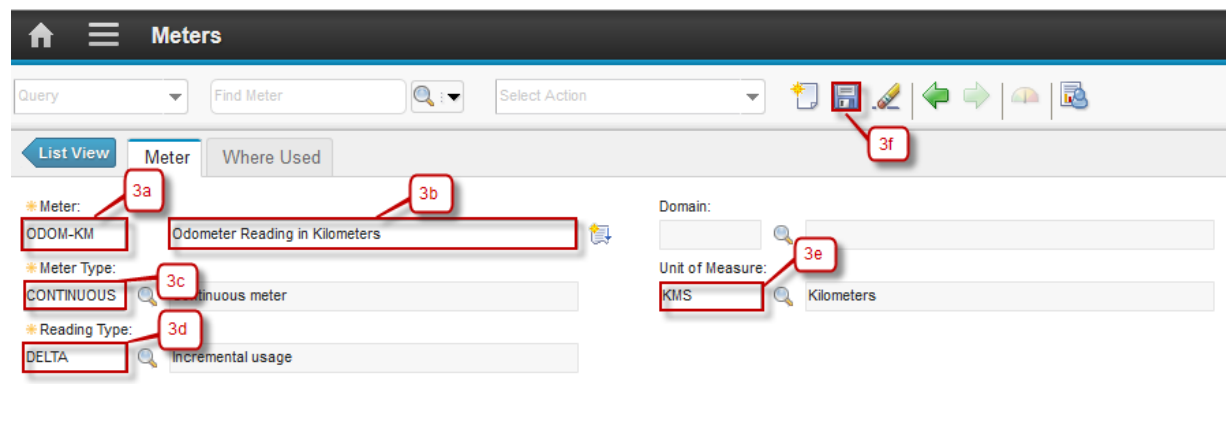
3c. Select the type of Meter.

3d. Input the Reading Type.

- If the type of Meter is 'CONTINUOUS', field Reading Type is mandatory.
- If the type of Meter is 'CHARACTERISTIC', field Domain is mandatory.

3e. Input the Unit of Measure

3f. Click **Save**.



The screenshot shows the 'Meters' form with the following fields and callouts:

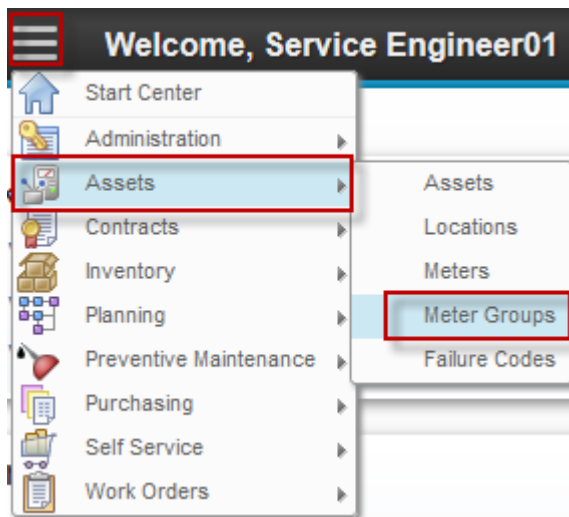
- 3a**: Points to the 'Meter' field containing 'ODOM-KM'.
- 3b**: Points to the 'Meter Description' field containing 'Odometer Reading in Kilometers'.
- 3c**: Points to the 'Meter Type' dropdown menu showing 'CONTINUOUS'.
- 3d**: Points to the 'Reading Type' dropdown menu showing 'DELTA'.
- 3e**: Points to the 'Unit of Measure' dropdown menu showing 'KMS'.
- 3f**: Points to the 'Save' button in the top right toolbar.

2.4.2 Create Meter Groups

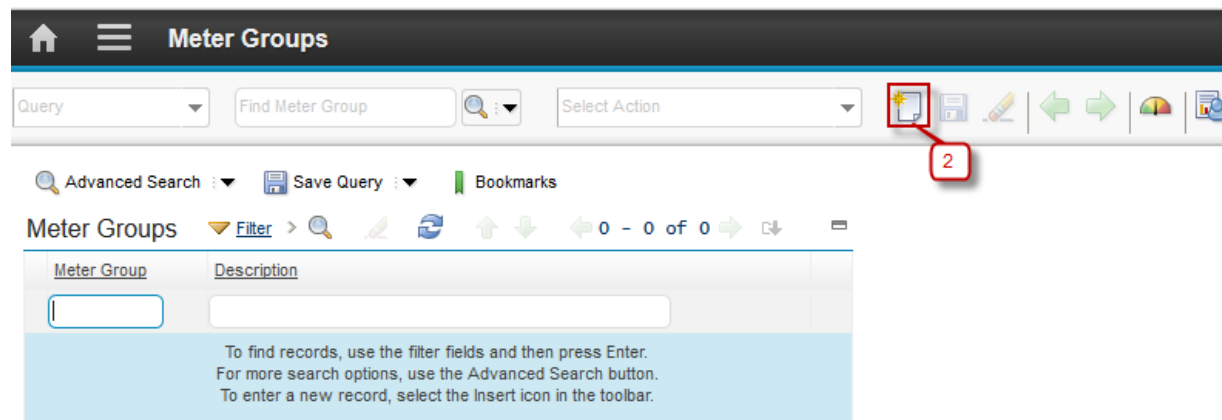
Below are step by step to Create Meter Groups by *Service Engineer*:

Step	Action
------	--------

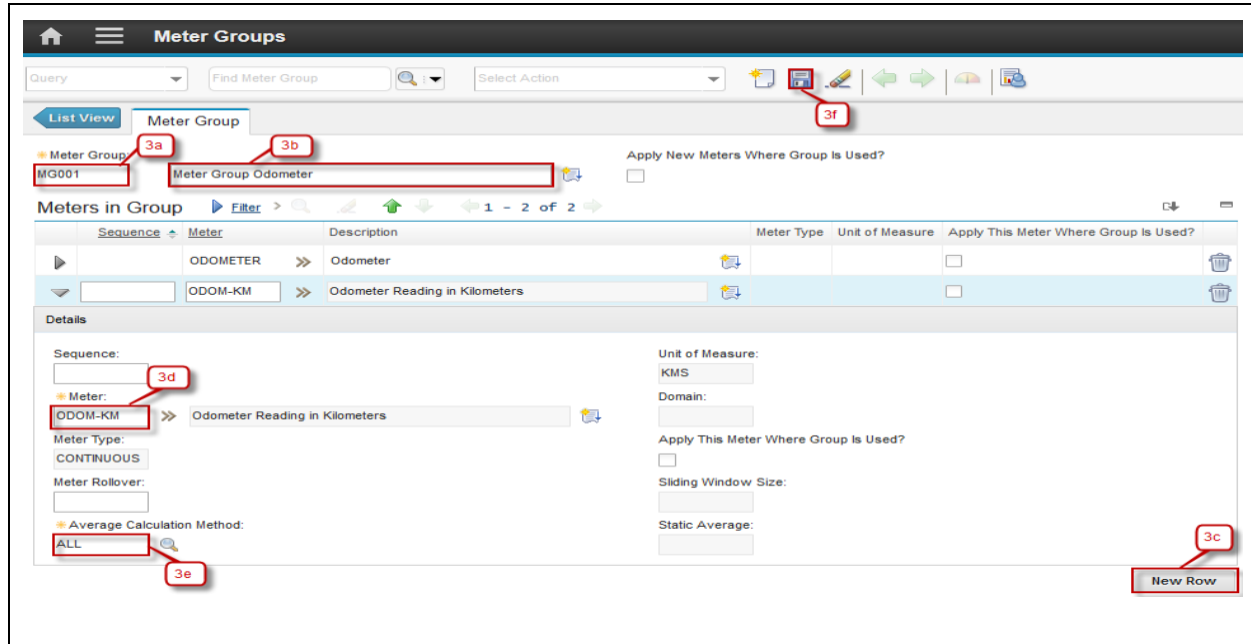
1. Go to Meters application by click  (GoTo) – Assets – Meter Groups.



2. Click  **New Row.**



- 3a. Input the Meter Group name.
- 3b. Input the Meter Group description.
- 3c. Click **New Row.**
- 3d. Select Meter using button 
- 3e. Select Average Calculation Method using button 
- 3f. Click **Save.**



The screenshot shows the 'Meter Groups' application interface. At the top, there is a navigation bar with a home icon, a menu icon, and the title 'Meter Groups'. Below this is a search bar with 'Find Meter Group' and a 'Select Action' dropdown. A toolbar contains icons for adding, editing, deleting, and other actions. The main content area is divided into two sections: 'List View' and 'Meter Group'. The 'List View' section shows a table of meter groups. The 'Meter Group' section shows the details for a selected meter group, 'MG001'. The details section includes fields for 'Sequence', 'Meter', 'Meter Type', 'Unit of Measure', 'Domain', 'Apply This Meter Where Group Is Used?', 'Sliding Window Size', 'Static Average', and 'Average Calculation Method'. Annotations 3a through 3f point to specific elements: 3a points to the 'Meter Group' dropdown, 3b points to the 'Meter Group Odometer' dropdown, 3c points to the 'New Row' button, 3d points to the 'Meter' dropdown, 3e points to the 'Average Calculation Method' dropdown, and 3f points to the 'Add' icon in the toolbar.

Sequence	Meter	Description	Meter Type	Unit of Measure	Apply This Meter Where Group Is Used?
	ODOMETER	Odometer			☐
	ODOM-KM	Odometer Reading in Kilometers			☐

Details

Sequence:

Meter: >> Odometer Reading in Kilometers

Meter Type: CONTINUOUS

Meter Rollover:

Average Calculation Method:

Unit of Measure: KMS

Domain:

Apply This Meter Where Group Is Used? ☐

Sliding Window Size:

Static Average:

New Row

2.5 Assets Application

Asset application is used to create and store asset numbers and corresponding information, such as parent, location, vendor, up/down status and maintenance costs.

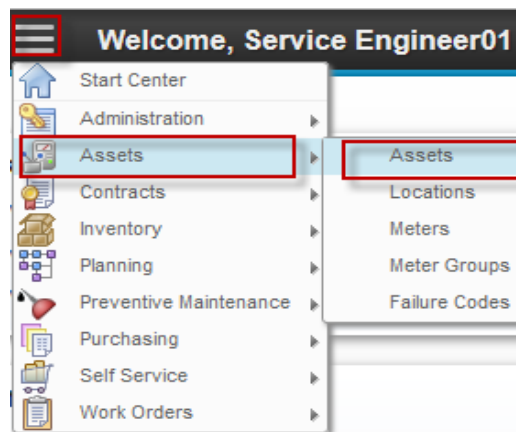
In **Asset** application user can build asset hierarchy, an arrangement of buildings, departments, assets and subassemblies. The asset hierarchy provides a convenient way to roll up maintenance costs so that user can check accumulated costs at any level, at any time. It also makes it easy to find a particular asset number.

2.5.1 Create Assets

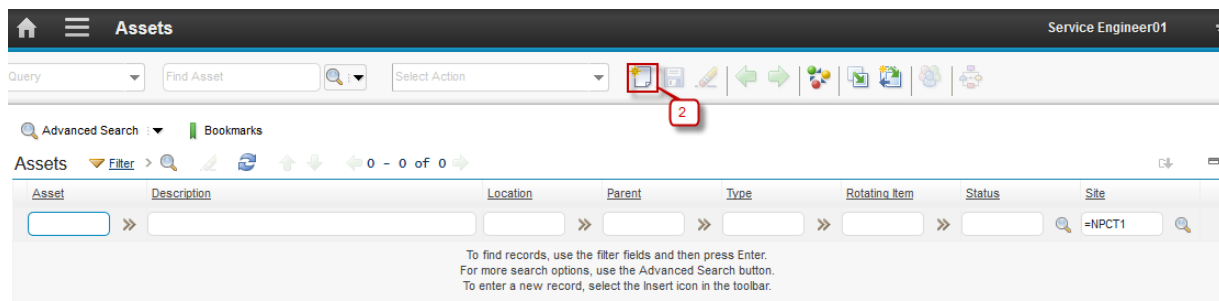
Asset creation can be done by *Service Engineer*. Below are step by step to Create Assets:

Step	Action
------	--------

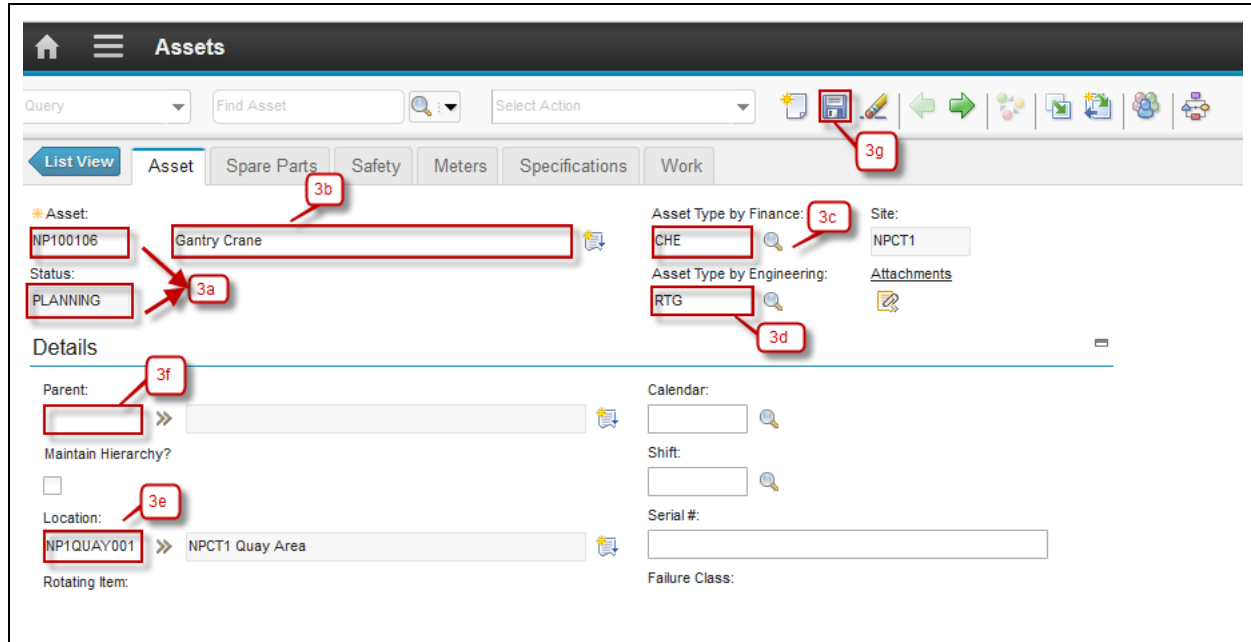
1. Go to Assets Application by click  (GoTo) – Assets – Assets.



2. Click  **New Row.**



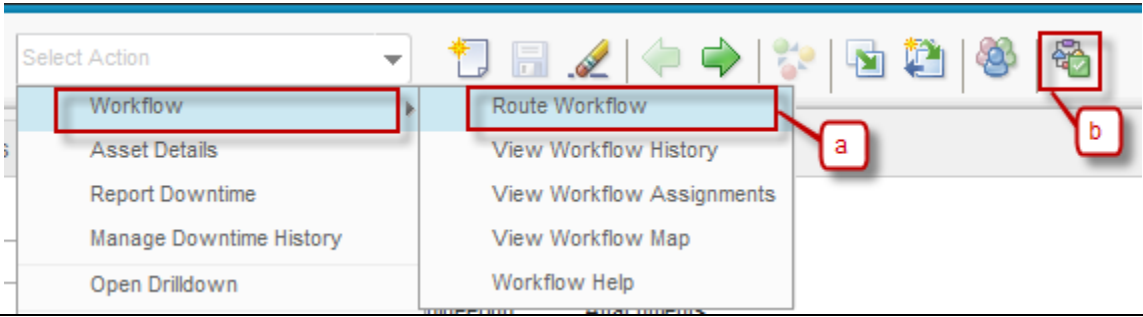
- 3a. Input the Asset name and make sure default status is 'PLANNING'.
- 3b. Input the Asset description.
- 3c. Select Asset Type by Finance using button .
- 3d. Select Asset Type by Engineering using button .
- 3e. Input Parent of Asset if the Asset have a parent.
- 3f. Input Location of Asset.
- 3g. Click **Save** and then after save the Asset button **Route Workflow** will be started automatically .



The screenshot shows the 'Assets' management interface. At the top, there's a 'Query' dropdown, a 'Find Asset' search bar, and a 'Select Action' dropdown. Below this is a tabbed interface with 'List View', 'Asset', 'Spare Parts', 'Safety', 'Meters', 'Specifications', and 'Work'. The 'Asset' tab is active. In the 'Asset' section, the 'Asset' field is 'NP100106', 'Status' is 'PLANNING', and 'Asset Type by Finance' is 'CHE'. The 'Details' section shows 'Parent' as 'NP1QUAY001' and 'Location' as 'NPCT1 Quay Area'. Callouts 3a through 3g point to various fields and buttons: 3a points to 'Status', 3b to 'Asset', 3c to 'Asset Type by Finance', 3d to 'Asset Type by Engineering', 3e to 'Location', 3f to 'Parent', 3g to the 'Save' button in the top right.

2.5.2 Asset Creation Approval by CHE Engineer

Created Asset must be approved by *CHE Engineer*. Below are step by step to Asset Creation Approval:

Step	Action
1. Asset Creation Approval have two ways:	
a. From Select Action → Workflow → Route Workflow	
b. Click button 	
	 The screenshot shows the 'Select Action' dropdown menu. The 'Workflow' option is highlighted, and its sub-menu is open, showing 'Route Workflow' as the first option. Callout 'a' points to 'Route Workflow' in the sub-menu, and callout 'b' points to the 'Route Workflow' button icon in the top right of the interface.
2. Click Route Workflow and dialog box will be appeared, see the figure below:	

Complete Workflow Assignment

Task:
Submit Asset NP100106 (Gantry Crane)

Action:
☒ Submit Asset
☐ Inactive

Memo:

Earlier Memos [Filter](#) >      0 - 0 of 0   

Memo	Person	Transaction Date
There are no rows to display.		

3a. Login as CHE Engineer 01.

3b. Assignment will be appeared in column **Inbox/Assignments** for approval Asset.

CHE Engineer 01

3a

Inbox / Assignments

Description	Last Memo	Start Date	Route
Next Assignment Due: 5/9/16 3:32 PM Refresh			
Need Approval for PR/ENG/10066/16/EMS (General)		2/15/16 2:07 PM	
Service Recovery 16000621 Needs to be Closed		2/23/16 9:04 PM	
Need Registration Approval for Asset NP100106 (Gantry Crane)		5/9/16 3:32 PM	

1 - 3 of 3

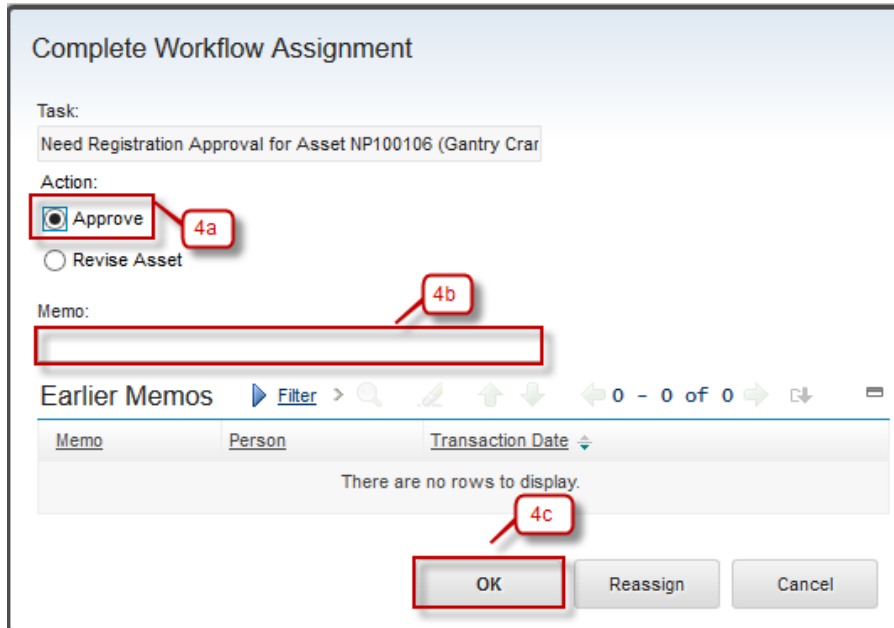
3b

4. Click button **Route Workflow**  and dialog box will be appeared see the figure below:

4a. Select Action Approve

4b. Input Memo (if needed)

4c. Click **OK**

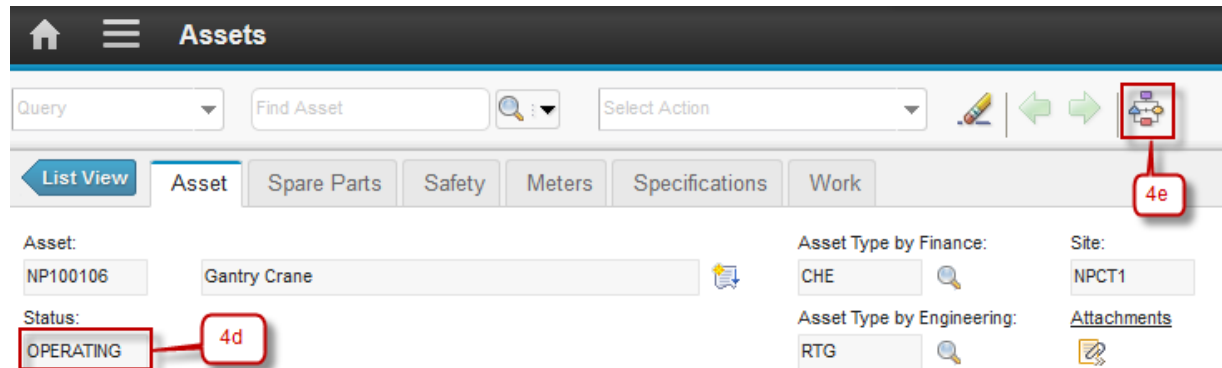


The dialog box titled "Complete Workflow Assignment" contains the following elements:

- Task:** A text field containing "Need Registration Approval for Asset NP100106 (Gantry Crar)".
- Action:** Two radio buttons: "Approve" (selected, labeled 4a) and "Revise Asset".
- Memo:** A text input field (labeled 4b) and a "Filter" button.
- Earlier Memos:** A table with columns "Memo", "Person", and "Transaction Date". It displays the message "There are no rows to display." (labeled 4c).
- Buttons:** "OK" (labeled 4c), "Reassign", and "Cancel".

4d. Status changed to 'OPERATING'.

4e. Workflow stopped.

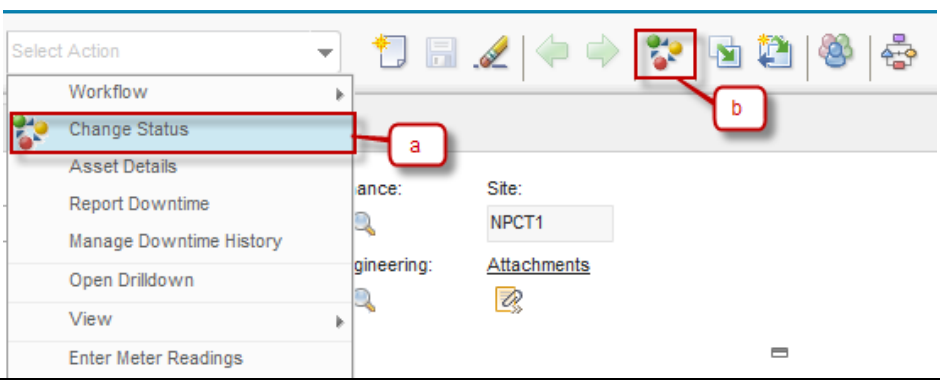
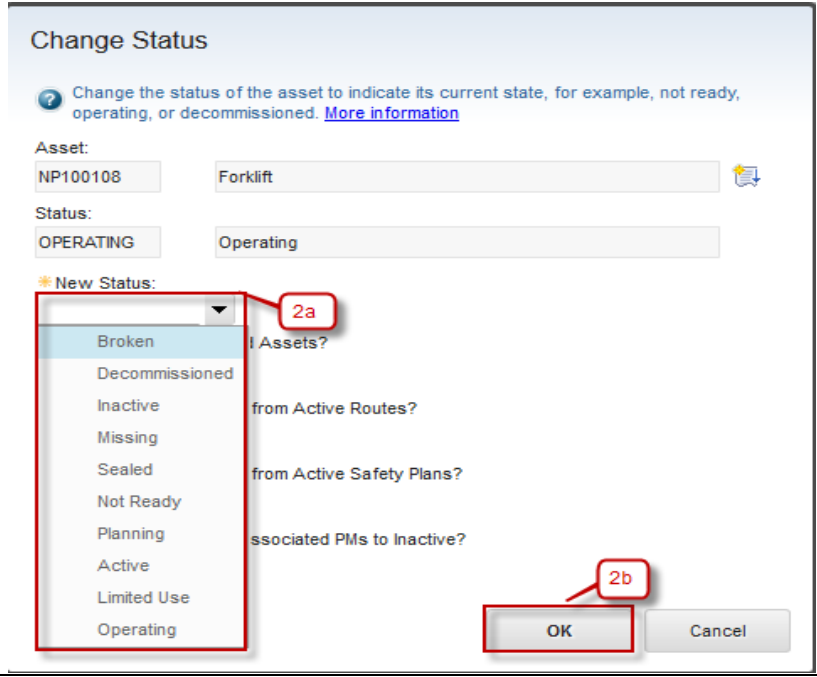


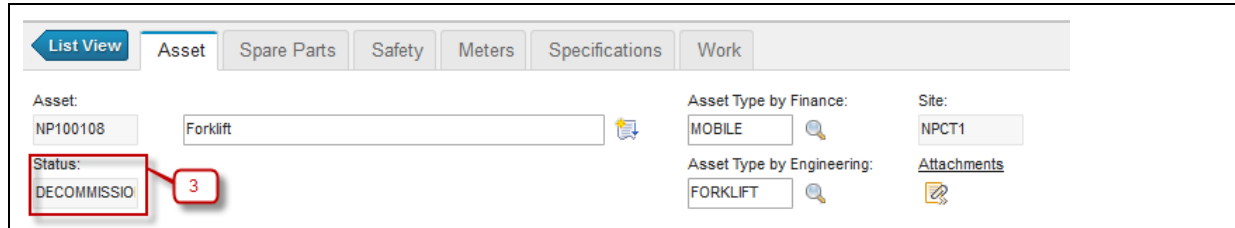
The "Assets" management interface shows the following details:

- Asset:** NP100106, Gantry Crane.
- Status:** OPERATING (labeled 4d).
- Asset Type by Finance:** CHE.
- Asset Type by Engineering:** RTG.
- Site:** NPCT1.
- Attachments:** A link to view attachments.
- Workflow:** A workflow icon (labeled 4e) is visible in the top right corner of the interface.

2.5.3 Change Status of an Asset

Sometimes, there is a needs to change the status of an asset. For example, when an asset is broken or decommissioned, user should change the status, to indicate that the asset is now in broken or decommissioned status. Below are step by step to Change Status of an Asset by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From Select Action → Change Status b. Click button 	
2. Dialog box Change Status will be appeared, see the figure below : 2a. Select New Status of Asset. 2b. Click OK .	
3. Status changed to DECOMMISSIONED	



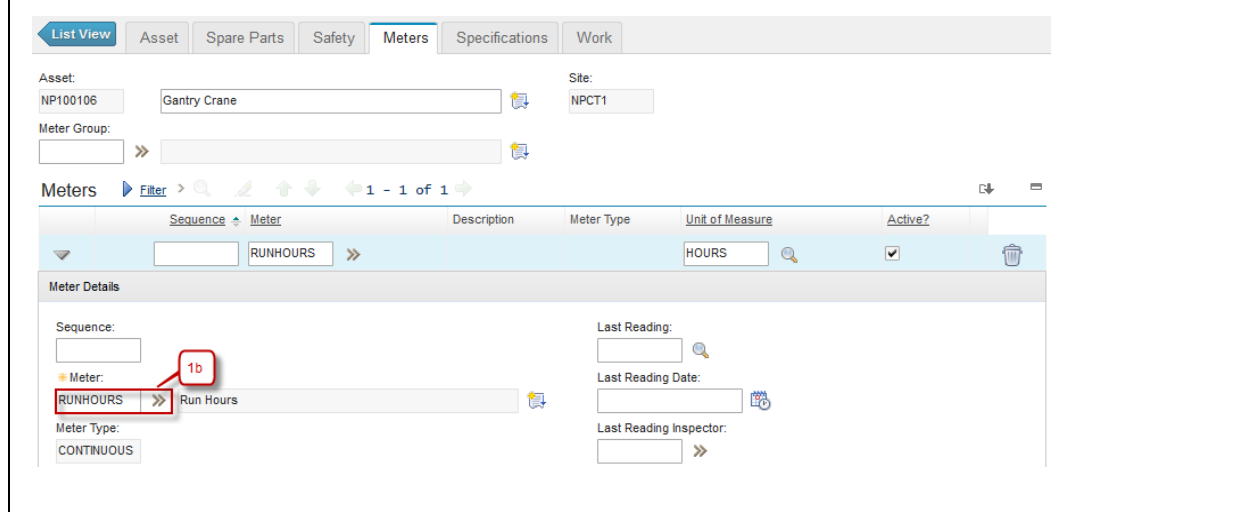
2.5.4 Assets Meters Registration & Meter Readings

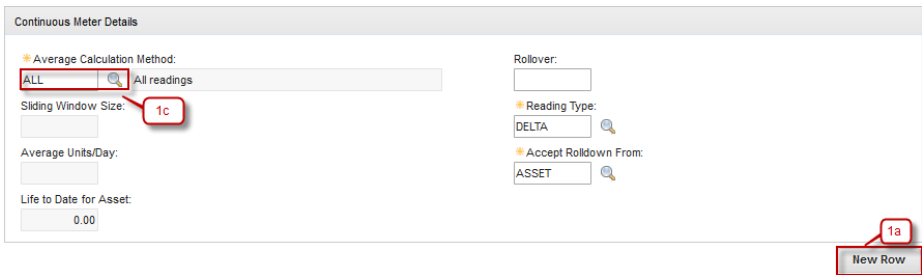
User can register all meters that apply to an asset in **Assets** application. Meters registration in an asset can be performed by registering a meter group that will automatically register all meters in meter group to the asset, or manually registering each meter to the asset.

In **Asset** application, user also can perform **Enter Meter Reading** action to report meter readings for the selected asset. This meter reading will be used as a measurement about the assets performance based the meters. This meter reading can be used to determine when a maintenance toward those asset should be performed. As additional notes, Enter Meter Reading action can be performed in **Work Order Tracking** application too (This application will be discussed later).

Below are step by step to register a meter to an asset, and performs **Enter Meter Reading** action, by *Service Engineer*:

Step	Action
1. Select one of Asset, go to Tab Meters .	
1a. Click New Row .	
1b. Select the Meter to use in Asset using button >> , field Meter Type and field Unit of Measure automatically fill.	
1c. If the type of Asset is 'CONTINUOUS' , field Average Calculation Method is mandatory.	





Continuous Meter Details

Average Calculation Method: **ALL** (1c) All readings

Sliding Window Size: 1c

Average Units/Day: 0.00

Life to Date for Asset: 0.00

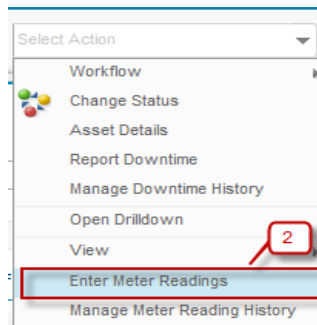
Rollover:

Reading Type: DELTA

Accept Rollover From: ASSET

1a New Row

2. To perform enter meter reading action, choose **Select Action** → **Enter Meter Readings**.



Select Action

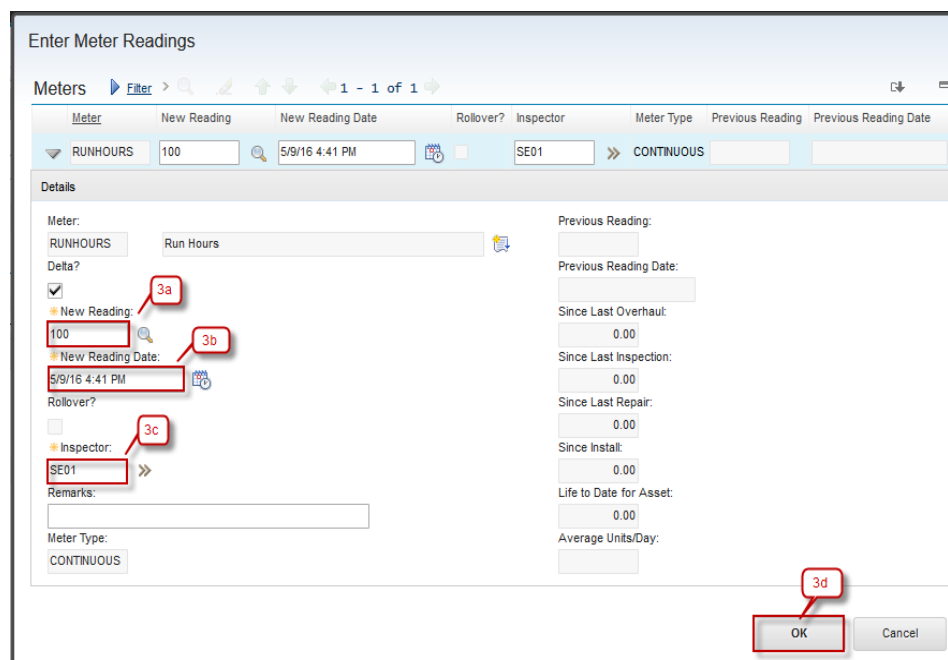
- Workflow
- Change Status
- Asset Details
- Report Downtime
- Manage Downtime History
- Open Drilldown
- View
- Enter Meter Readings** (2)
- Manage Meter Reading History

3a. Input value New Reading.

3b. New Reading Date automatically fill when input New Reading.

3c. Inspector automatically fill with user login.

3d. Click **OK**.



Enter Meter Readings

Meters Filter 1 - 1 of 1

Meter	New Reading	New Reading Date	Rollover?	Inspector	Meter Type	Previous Reading	Previous Reading Date
RUNHOURS	100	5/9/16 4:41 PM		SE01	CONTINUOUS		

Details

Meter: RUNHOURS Run Hours

Delta? ☒ 3a

New Reading: 100 3b

New Reading Date: 5/9/16 4:41 PM

Rollover? ☐ 3c

Inspector: SE01

Remarks:

Meter Type: CONTINUOUS

Previous Reading:

Previous Reading Date:

Since Last Overhaul: 0.00

Since Last Inspection: 0.00

Since Last Repair: 0.00

Since Install: 0.00

Life to Date for Asset: 0.00

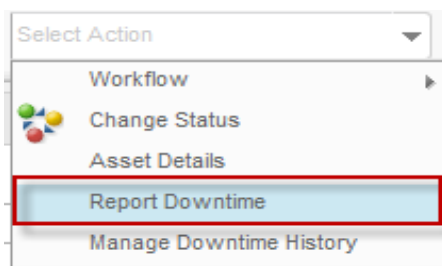
Average Units/Day:

3d OK Cancel

2.5.5 Downtime

Downtime is another feature that available in an asset. This feature can be used to determine and track how long an asset has been out of service. With this feature, user can be informed about how effective an asset is. A field of total accumulated downtime that ever happens to the asset is available in **Asset** application too.

Downtime is reported in **Assets** application using **Report Downtime** action. In this action, user should define how long a downtime occurs to the asset, the start time, and the end time. As a note, this report downtime feature also available in **Work Order Tracking** application. Below are step by step to report downtime of an asset:

Step	Action
1. Select one of Asset. Select Action → Report Downtime	

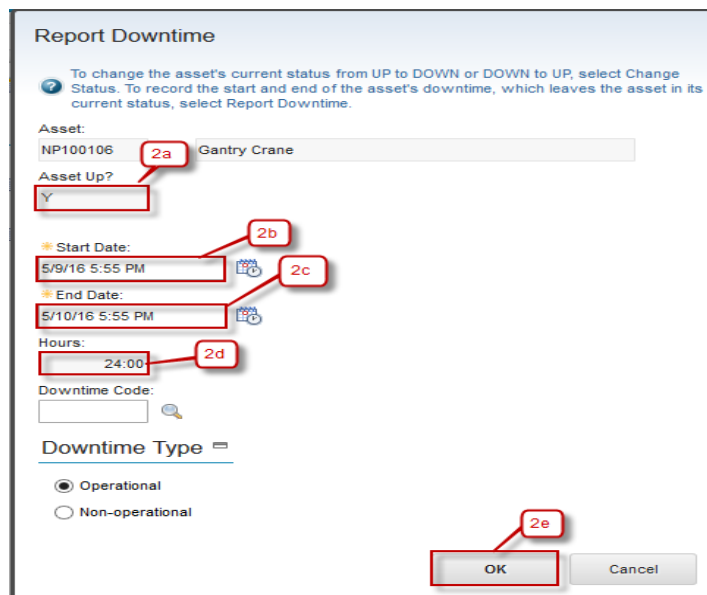
2a. Make sure default **Asset Up?** is **Y**.

2b. Start Date usually automatically fill with current date, but it's editable.

2c. Input End Date until when the asset will be up again.

2d. Value of Hours is interval between Start Date and End Date.

2e. Click **OK**.



Report Downtime

To change the asset's current status from UP to DOWN or DOWN to UP, select Change Status. To record the start and end of the asset's downtime, which leaves the asset in its current status, select Report Downtime.

Asset: NP100106 **2a** Gantry Crane

Asset Up? ☒ Y

Start Date: 5/9/16 5:55 PM **2b**

End Date: 5/10/16 5:55 PM **2c**

Hours: 24:00 **2d**

Downtime Code:

Downtime Type

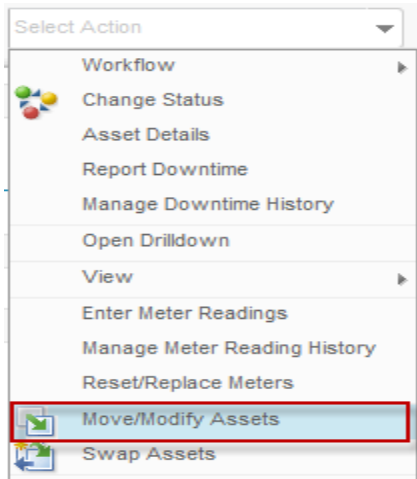
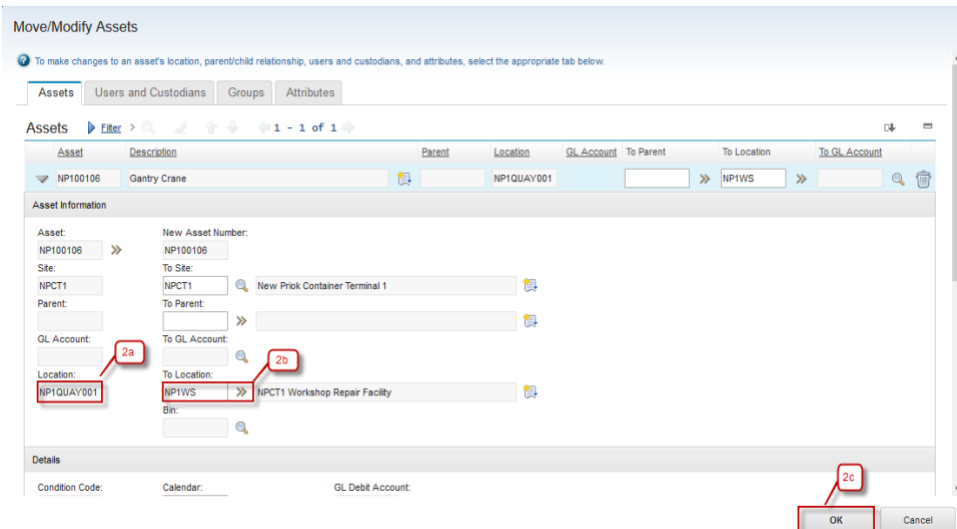
☒ Operational

☐ Non-operational

2e OK Cancel

2.5.6 Move Assets

When an asset need to be moved from a location to another location, for example, from operating location to repair location, user can do that from **Asset** application via **Move/Modify Assets** action. Besides locations, parent asset and condition code (for rotating asset) can be changed through this action too. Below are step by step to move location of an asset:

Step	Action
1. Select one of Asset. Select Action → Move/Modify Assets	
2a. Current Location before being moved. 2b. Field To Location where the Asset move. 2c. Click OK . 2d. Asset already move to New Location.	

[List View](#) | **Asset** | Spare Parts | Safety | Meters | Specifications | Work

Asset:
NP100106

Gantry Crane

Status:
OPERATING

Asset Type by Finance:
CHE

Asset Type by Engineering:
RTG

Site:
NPCT1

[Attachments](#)

Details

Parent:
»

Maintain Hierarchy?
☐

Location:
NP1WS » NPCT1 Workshop Repair Facility

Rotating Item:
»

Meter Group:
»

Calendar:

Shift:

Serial #:

Failure Class:
»

2.6 Exercise Phase